Part 3: Analysis of Variance (ANOVA)

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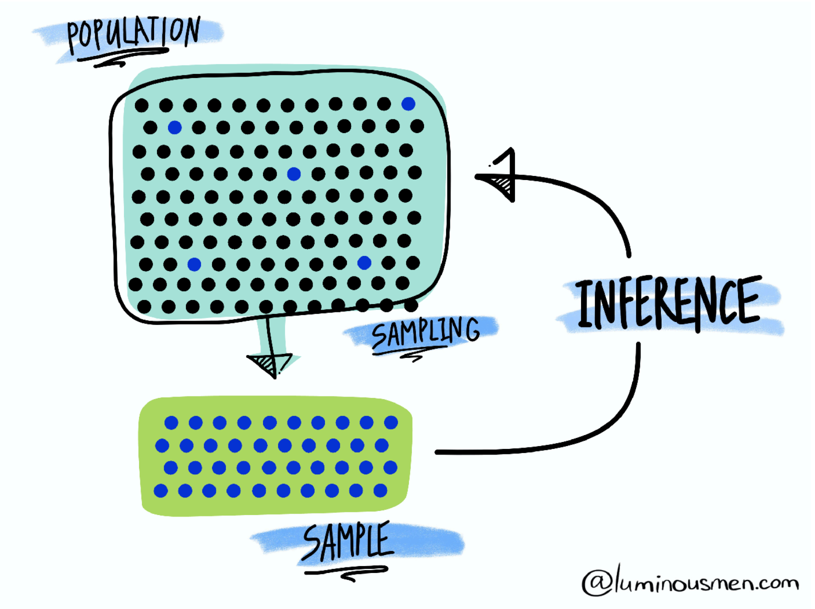
How does it work?

Statistical methods are based on several fundamental concepts, the most central of which is to consider the information available (in the form of data) resulting from a random process.

As such, the data represent a random sample of a totally or conceptually accessible population.

Then, statistical inference allows to infer the properties of a population based on the observed sample. This includes deriving estimates and testing hypotheses.



Source: luminousmen

Hypothesis testing

- In general (scientific) hypotheses can be translated into a set of (non-overlapping idealized) statistical hypotheses:

$$H_0 : \theta \in \Theta_0 \quad \text{and} \quad H_a : \theta \notin \Theta_0$$

- In a hypothesis test, the statement being tested is called the null hypothesis H_0. A hypothesis test is designed to assess the strength of the evidence against the null hypothesis.
- The alternative hypothesis H_a is the statement we hope or suspect to be true instead of H_0.
- Each hypothesis excludes the other, so that one can exclude one in favor of the other using the data.
- Example: a drug represses the progression of cancer

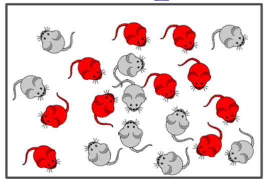
$$H_0 : \mu_{\text{drug}} = \mu_{\text{control}} \quad \text{and} \quad H_a : \mu_{\text{drug}} < \mu_{\text{control}}.$$

Hypothesis testing

Statistical Hypotheses

H_0: $\mu_{Drug} = \mu_{Control}$
H_a: $\mu_{Drug} < \mu_{Control}$

Sample



Data: Tumour size at day 42

$(x_{D,1}; x_{D,2}; ...; x_{D,n_D})$
$(x_{C,1}; x_{C,2}; ...; x_{C,n_C})$

Point estimation

$\widehat{\mu}_{Drug} - \widehat{\mu}_{Control}$

Inference: Under H0

$$T_{obs} = \frac{\widehat{\mu}_{Drug} - \widehat{\mu}_{Control}}{s_p \sqrt{\frac{1}{n_D} + \frac{1}{n_C}}} \sim St_{n_D + n_C - 2}$$

Idea:
Drug represses the progression of cancer

Hypothesis testing

Outcome	H_0 **is true**	H_0 **is false**
Can't reject H_0	✅ Correct decision (prob=$1-\alpha$)	⚠️ Type II error (prob=$1-\beta$)
Reject H_0	⚠️ Type I error (prob=α)	✅ Correct decision (prob=β)

- The type I error corresponds to the probability of rejecting H_0 when H_0 is true (also called false positive). The type II error corresponds to the probability of not rejecting H_0 when H_a is true (also called false negative).
- A test is of significance level α when the probability of making a type I error equals α. Usually we consider $\alpha = 5\%$, however, this can vary depending on the context.
- A test is of power β when the probability to make a type II error is $1-\beta$. In other words, the power of a test is its probability of rejecting H_0 when H_0 is false (or the probability of accepting H_a when H_a is true).

What are p-values?

In statistics, the **p-value** is defined as the probability of observing a test statistic that is at least as extreme as actually observed, assuming that the null hypothesis H_0 is true.

Informally, a p-value can be understood as a measure of plausibility of the null hypothesis given the data. A small p-value indicates strong evidence against H_0.

When the p-value is small enough (i.e., smaller than the significance level α), the test based on the null and alternative hypotheses is considered significant, meaning we reject the null hypothesis in favor of the alternative. This is generally what we want because it "verifies" our (research) hypothesis.

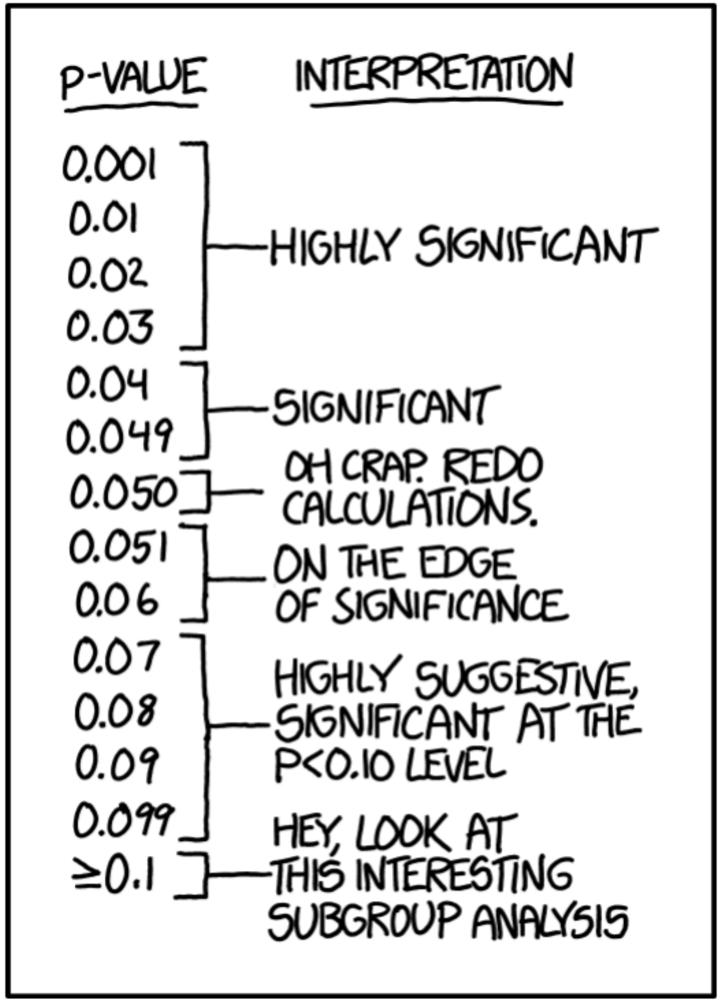
When the p-value is not small enough, with the available data, we cannot reject the null hypothesis, so nothing can be concluded. 🤔

The obtained p-value summarizes the incompatibility between the data and the model constructed under the set of assumptions.

"Absence of evidence is not evidence of absence." 👋

👋 From the British Medical Journal.

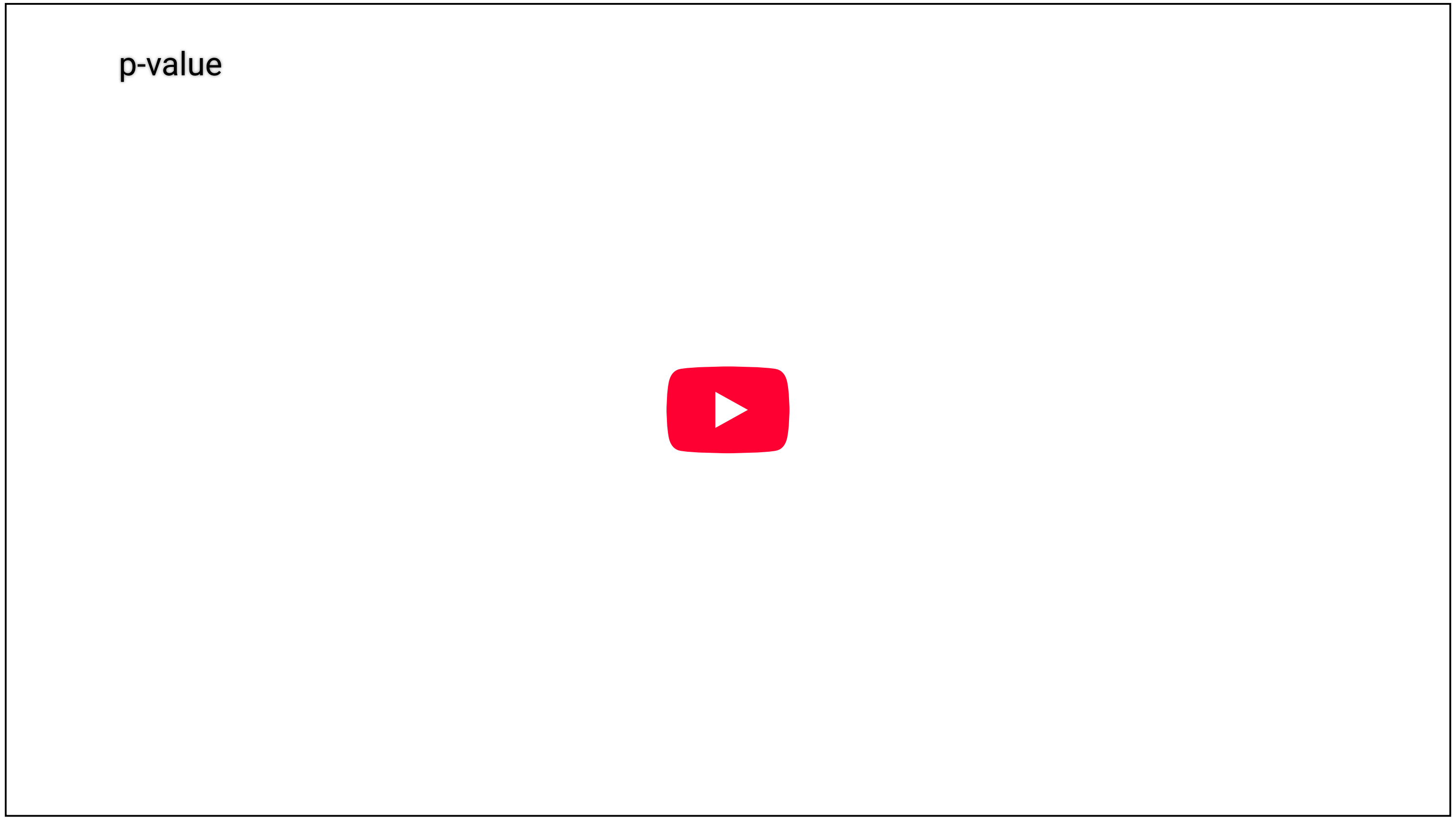
How to understand p-values?



👋 If you want to know more have a look here.

P-values may be controversial

P-values have been misused many times because understanding what they mean is not intuitive!



👋 If you want to know more have a look here.

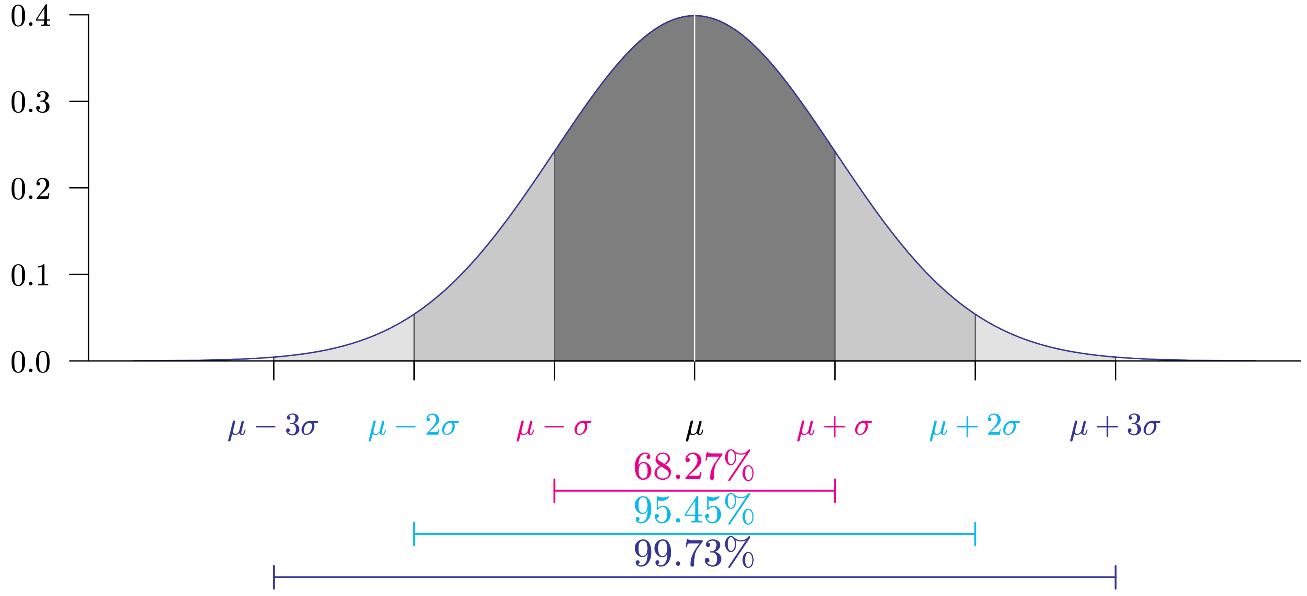
Quick review: Normal distribution

$$Y \sim N(\mu, \sigma^2), \qquad f_Y(y) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{(y-\mu)^2}{2\sigma^2}}$$

$$\mathbb{E}[Y] = \mu, \qquad \mathrm{Var}[Y] = \sigma^2,$$

$$Z = \frac{Y-\mu}{\sigma} \sim N(0,1), \qquad f_Z(z) = \frac{1}{\sqrt{2\pi}}\, e^{-\frac{z^2}{2}}.$$

Probability density function of a normal distribution:



Two-sample location tests

In practice, we often encounter problems where our goal is to compare the means (or locations) of two samples. For example,

1. A scientist is interested in comparing the vaccine efficacy of the Pfizer-BioNTech and the Moderna vaccine.
2. A bank wants to know which of two proposed plans will most increase the use of its credit cards.
3. A psychologist wants to compare male and female college students' impression on a selected webpage.

We will discuss three two-sample location tests:

1. Two independent sample Student's t-test
2. Two independent sample Welch's t-test
3. Two independent sample Mann-Whitney-Wilcoxon test

Two independent sample Student's t-test

This test considers the following assumed model for group **A** and **B**

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)} = \mu + \delta_g + \varepsilon_{i(g)},$$

where $g = A, B$, $i = 1, \ldots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2)$ and $\sum n_g \delta_g = 0$.

📝 n_A = sample size of group **A**, $\mu_A = \mu + \delta_A$ = population mean of group **A**, n_B and $\mu_B = \mu + \delta_B$ are similarly defined for group **B**.

Hypotheses:

$$H_0 : \mu_A - \mu_B = \mu_0 \quad \text{and} \quad H_a : \mu_A - \mu_B \; [> \text{ or } < \text{ or } \neq] \, \mu_0.$$

Test statistic's distribution under H_0:

$$T = \frac{(\overline{X}_A - \overline{X}_B) - \mu_0}{s_p \sqrt{n_A^{-1} + n_B^{-1}}} \underset{H_0}{\sim} \text{Student}(n_A + n_B - 2).$$

Discussion — Student's t-test

Python function:

```
from scipy import stats

stats.ttest_ind(a = ..., b = ..., alternative = ..., equal_var = True)
```

This test strongly relies on the assumed absence of outliers. If outliers appear to be present the Mann-Whitney-Wilcoxon test (see later) is (probably) a better option.

For moderate and small sample sizes, the sample distribution should be at least approximately normal with no strong skewness to ensure the reliability of the test.

In practice, the assumption of equal variance is hard to verify so **we recommend to avoid this test in practice**.

Two independent sample Welch's t-test

This test considers the following assumed model for group **A** and **B**

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)} = \mu + \delta_g + \varepsilon_{i(g)},$$

where $g = A, B$, $i = 1, \ldots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma_g^2)$ and $\sum n_g \delta_g = 0$.

📝 n_A = sample size of group **A**, $\mu_A = \mu + \delta_A$ = population mean of group **A**, n_B and $\mu_B = \mu + \delta_B$ are similarly defined for group **B**.

Hypotheses:

$$H_0 : \mu_A - \mu_B = \mu_0 \quad \text{and} \quad H_a : \mu_A - \mu_B \; \big[> \text{ or } < \text{ or } \neq \big] \mu_0.$$

Test statistic's distribution under H_0:

$$T = \frac{(\overline{X}_A - \overline{X}_B) - \mu_0}{\sqrt{s_A^2/n_A + s_B^2/n_B}} \underset{H_0}{\sim} \text{Student}(df).$$

Discussion — Welch's t-test

Python function:

```
from scipy import stats

stats.ttest_ind(a = ..., b = ..., alternative = ..., equal_var = False)
```

This test strongly relies on the assumed absence of outliers. If outliers appear to be present the Mann-Whitney-Wilcoxon test (see later) is (probably) a better option.

For moderate and small sample sizes, the sample distribution should be at least approximately normal with no strong skewness to ensure the reliability of the test.

This test does not require the variances of the two groups to be equal. If the variances of the two groups are the same (which is rather unlikely in practice), the Welch's t-test loses a little bit of power compared to the Student's t-test.

The computation of df (i.e. the degrees of freedom of the distribution under the null) is beyond the scope of this class.

Mann-Whitney-Wilcoxon test

This test considers the following assumed model for group **A** and **B**

$$X_{i(g)} = \theta_g + \varepsilon_{i(g)} = \theta + \delta_g + \varepsilon_{i(g)},$$

where $g = A, B$, $i = 1, \dots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2)$ and $\sum n_g \delta_g = 0$.

📝 n_A = sample size of group **A**, $\theta_A = \theta + \delta_A$ = population mean of group **A**, n_B and $\theta_B = \theta + \delta_B$ are similarly defined for group **B**.

Hypotheses:

$$H_0 : \theta_A - \theta_B = \theta_0 \quad \text{and} \quad H_a : \theta_A - \theta_B \; \left[> \text{ or } < \text{ or } \neq\right] \theta_0.$$

Test statistic's distribution under H_0:

$$Z = \frac{\sum_{i=1}^{n_B} R_{i(g)} - [n_B(n_A + n_B + 1)/2]}{\sqrt{n_A n_B (n_A + n_B + 1)/12}},$$

where $R_{i(g)}$ denotes the global rank of the i-th observation of group g.

Discussion – Mann–Whitney–Wilcoxon

Python function:

```
from scipy import stats

stats.ranksums(a = ..., b = ...)
```

This test is “robust” in the sense that (unlike the t-tests) it is not overly affected by outliers.

For the Mann–Whitney–Wilcoxon test to be comparable to the t-tests (i.e. testing for the mean) we need to assume **symmetric distributions** and **equality in variances** .

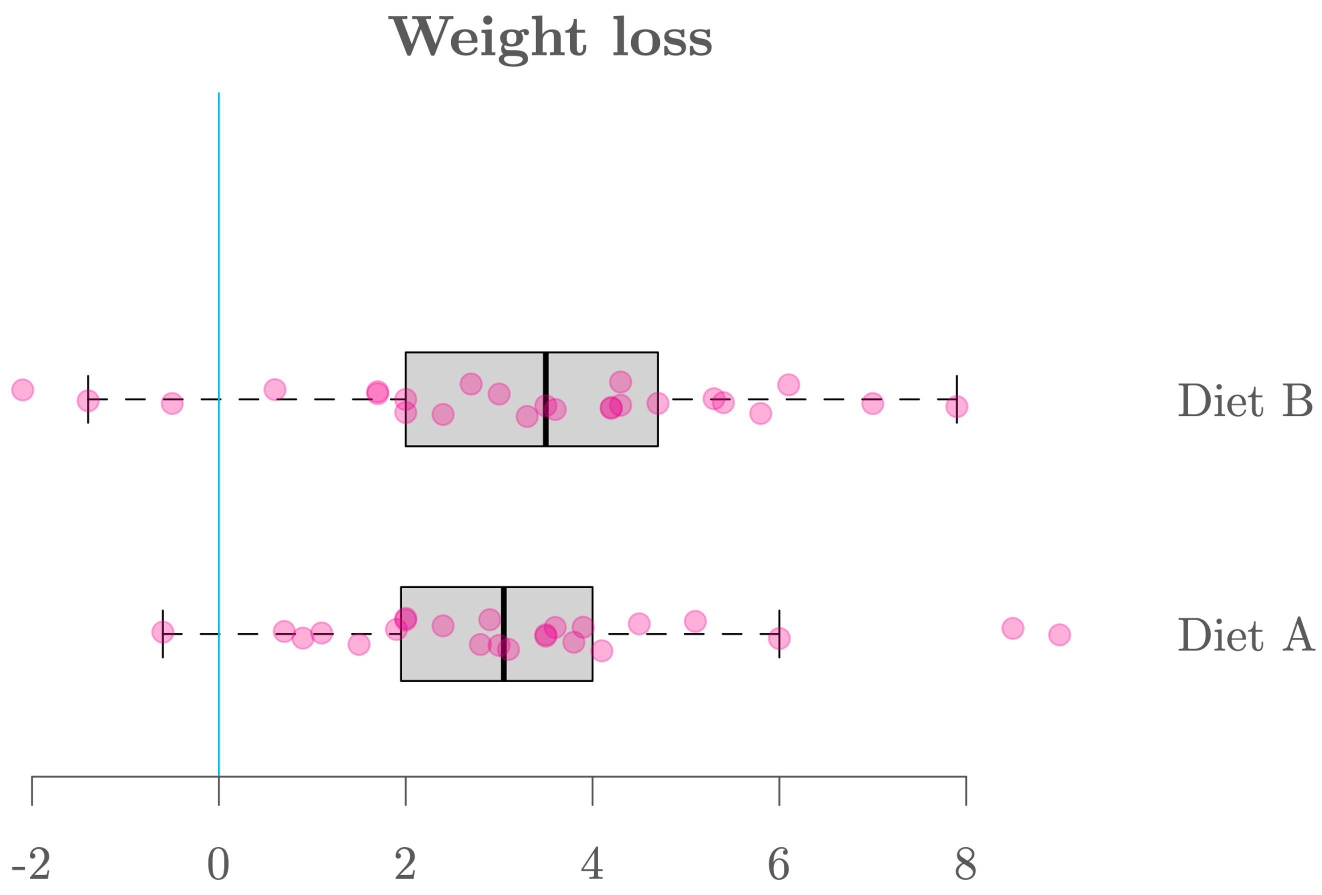
Then, we have $\theta_A = \mu_A$ and $\theta_B = \mu_B$.

Compared to the t-tests, the Mann–Whitney–Wilcoxon test is less powerful if their requirements (Gaussian and possibly same variances) are met.

The distribution of this method under the null is complicated and can be obtained by different methods (e.g. exact, asymptotic normal, ...).
The details are beyond the scope of this class.

Comparing diets A and B

Graph



Import

```
# Import data
import pandas as pd
diet = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data/diet.
n = 5
print(diet.head(n)) #print the first n rows of the dataset
```

```
   id  gender  age  height diet.type  initial.weight  final.weight
0   1  Female   22     159         A              58          54.2
1   2  Female   46     192         A              60          54.0
2   3  Female   55     170         A              64          63.3
3   4  Female   33     171         A              64          61.1
4   5  Female   50     170         A              65          62.2
```

```

# Compute weight loss
diet["weight.loss"] = diet["initial.weight"] - diet["final.weight"]

# Select diet
posA = diet["diet.type"] == "A"
posB = diet["diet.type"] == "B"

# Variable of interest
dietA = diet["weight.loss"][diet["diet.type"]=="A"]
dietB = diet["weight.loss"][diet["diet.type"]=="B"]
```

Student

```python
from scipy.stats import ttest_ind

t_stat, p_val = ttest_ind(dietA, dietB, alternative="two-sided", equal_var=True)

print("t-statistic:", round(t_stat, 4))
```

```
t-statistic: 0.0475
```

```python
print("p-value:", round(p_val, 4))
```

```
p-value: 0.9623
```

This document was prepared with the help of Lionel Voirol and Filippo Salmaso

Welch

```python
from scipy import stats

# Welch's t-test (equal_var=False makes it Welch)
t_stat, p_value = stats.ttest_ind(dietA, dietB, equal_var=False)

print("t-statistic:", round(t_stat, 4))
```

```
t-statistic: 0.0476
```

```python
print("p-value:", round(p_value, 4))
```

```
p-value: 0.9622
```

Wilcox

```python
from scipy import stats

# Wilcoxon rank-sum test (Mann–Whitney U)
stat, p_value = stats.ranksums(dietA, dietB)

print("Wilcoxon statistic:", round(stat, 4))
```

```
Wilcoxon statistic: -0.46
```

```python
print("p-value:", round(p_value, 4))
```

```
p-value: 0.6455
```

Results

Define hypotheses: $H_0 : \mu_A = \mu_B, \quad H_a : \mu_A \neq \mu_B$

Define

We consider $\alpha = 5\%$

Compute p-value

Welch's *t*-test appears suitable in this case, and therefore, we obtain $\text{p-value} = 96.22\%$ (see R output tab for details).

Conclusion

We have $\text{p-value} > \alpha$ so we **fail to reject the null hypothesis** at the significance level of 5%.

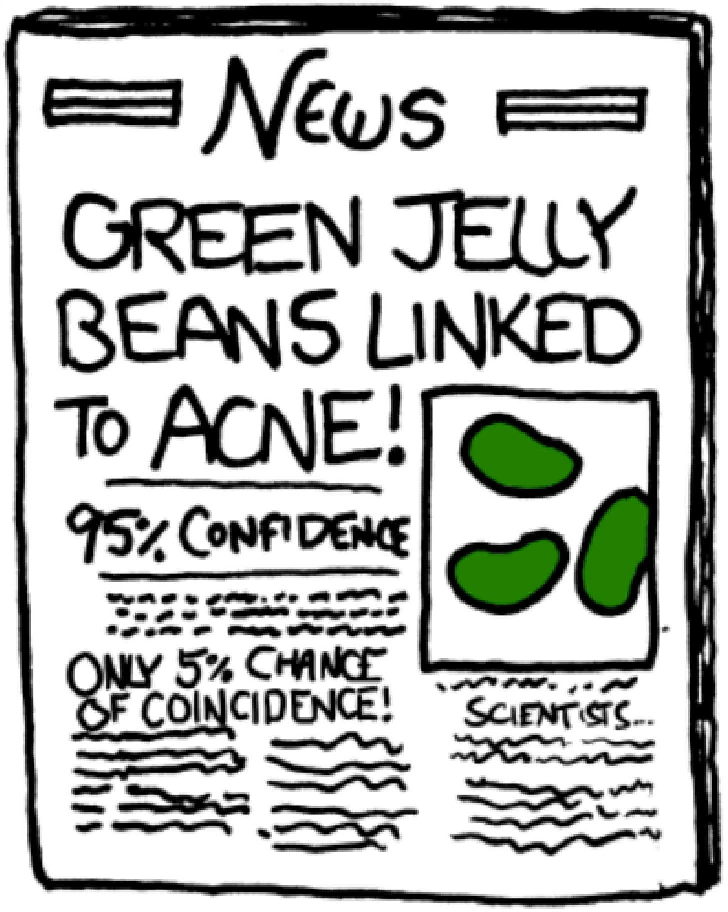
Problems with multiple samples

In practice, we often even encounter situations where we need to compare the means of more than 2 groups. For example, we want to compare the weight loss efficacy of several diets, say diets **A**, **B**, **C**. Your theory could, for example, be the following: $0 < \mu_A = \mu_B < \mu_C$. A possible approach to evaluate its validity:

1. **Show that μ_C is greater than μ_A and μ_B** (Test 1: $H_0 : \mu_A = \mu_C$, $H_a : \mu_A < \mu_C$; Test 2: $H_0 : \mu_B = \mu_C$, $H_a : \mu_B < \mu_C$). Here we hope to **reject H_0 in both cases**.
2. **Show that μ_A and μ_B are greater than 0** (Test 3: $H_0 : \mu_A = 0$, $H_a : \mu_A > 0$; Test 4: $H_0 : \mu_B = 0$, $H_a : \mu_B > 0$). Here we also hope to **reject H_0 in both cases**.
3. **Compare μ_A and μ_B** (Test 5: $H_0 : \mu_A = \mu_B$, $H_a : \mu_A \neq \mu_B$). Here we hope **not to reject H_0**. ⚠️ This does **not imply** that $\mu_A = \mu_B$ is true, but at least the result would not contradict our theory.

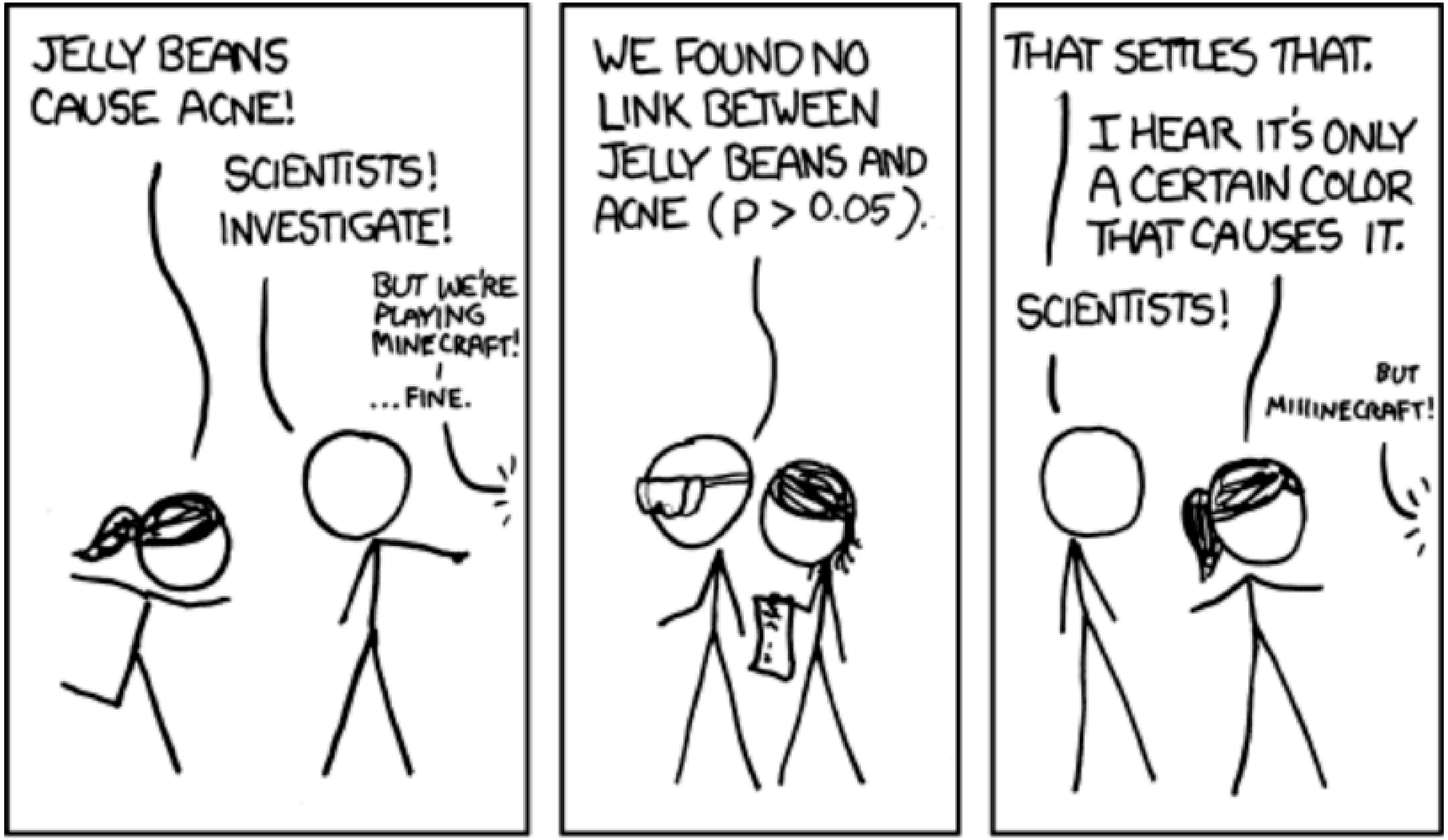
Is there a problem in doing many tests?

Are jelly beans causing acne? Maybe... but why only **green** ones? 🤨



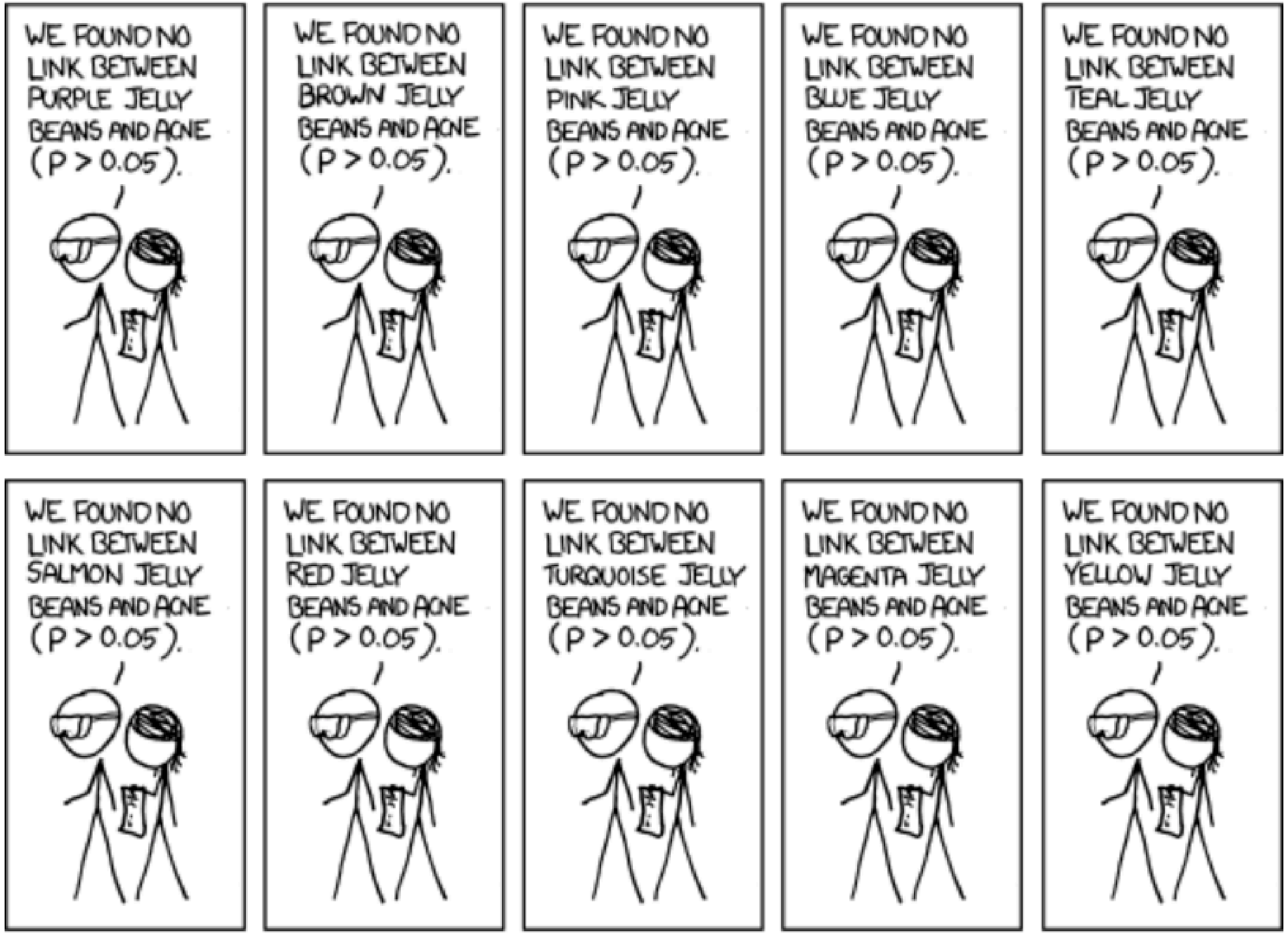
Source: xkcd

Are jelly beans causing acne?



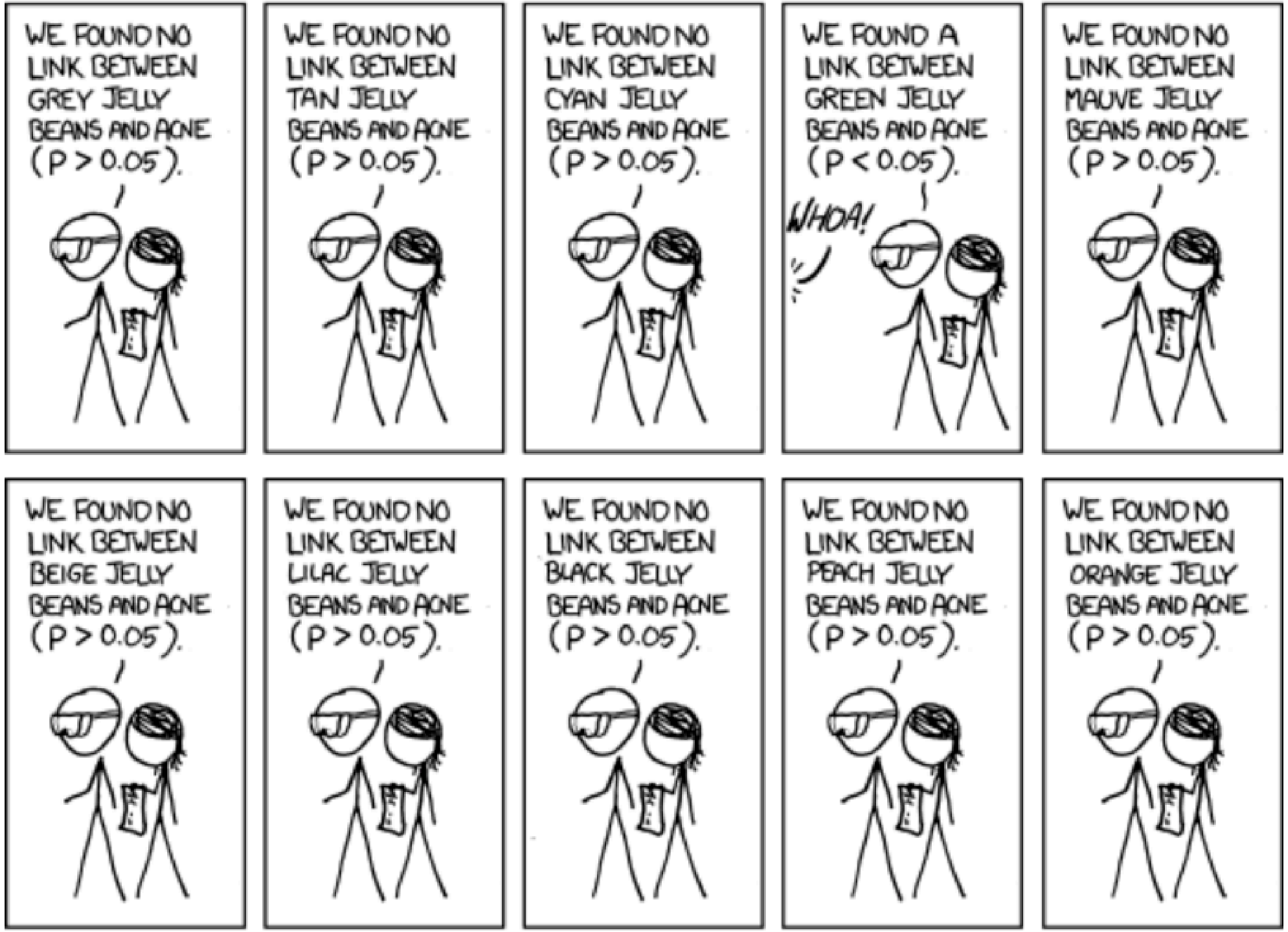
Source: xkcd

Maybe a specific color?



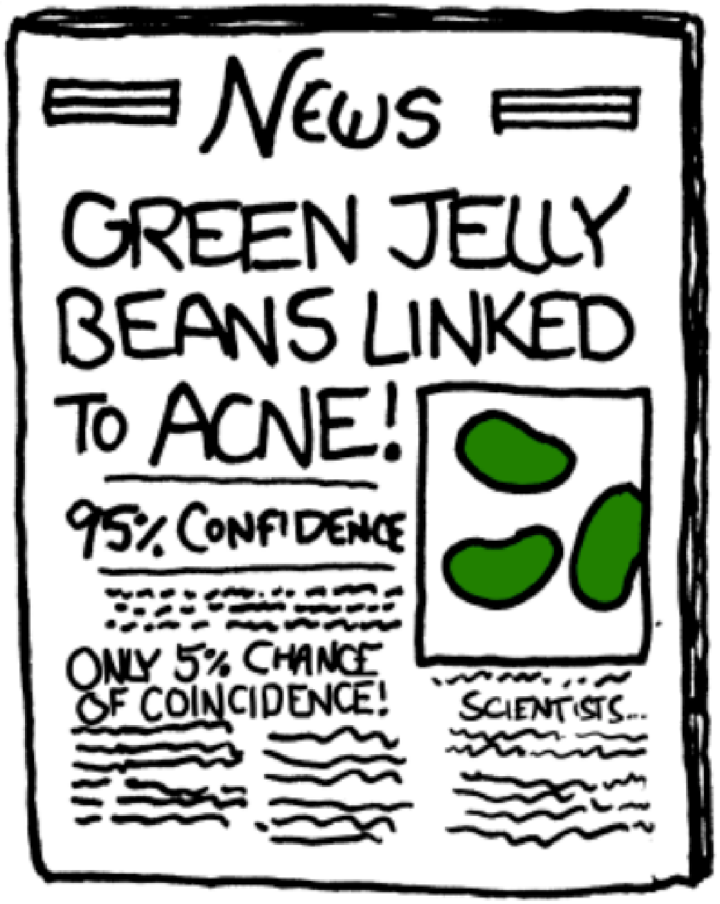
Source: xkcd

Maybe a specific color?



Source: xkcd

And finally...



Source: xkcd

👋 If you want to know more about this comic, have a look here.

Multiple testing can be dangerous!

- Remember that a p-value is random as its value depends on the data.
- If multiple hypotheses are tested, the chance of observing a rare event increases, and therefore, the chance to incorrectly reject a null hypothesis (i.e. making a Type I error) increases.
- For example, if we consider k (independent) tests (whose null hypotheses are all correct), we have

$$\begin{aligned}\alpha_k &= \Pr(\text{reject } H_0 \text{ at least once}) \\ &= 1 - \Pr(\text{not reject } H_0 \text{ test } 1) \times \ldots \times \Pr(\text{not reject } H_0 \text{ test k}) \\ &= 1 - (1-\alpha) \times \ldots \times (1-\alpha) = 1 - (1-\alpha)^k\end{aligned}$$

- Therefore, α_k increases rapidly with k (e.g. $\alpha_1 = 0.05$, $\alpha_2 \approx 0.098$, $\alpha_{10} \approx 0.4013$, $\alpha_{100} \approx 0.9941$).
- Hence performing multiple tests, with the same or different data, is dangerous ⚠️ (but very common! 😟) as it can lead to significant results, when actually there are none!

Possible solutions

Suppose that we are interested in making k tests and that we want the probability of rejecting the null at least once (assuming the null hypotheses to be correct for all tests) α_k to be equal to α (typically 5%). Instead of using α for the individual tests we will use α_c (i.e. a corrected α). Then, for k (potentially dependent) tests we have

$$\begin{aligned}
\alpha_k = \alpha &= \Pr(\text{reject } H_0 \text{ at least once}) \\
&= \Pr(\text{reject } H_0 \text{ test 1 OR } \dots \text{ OR reject } H_0 \text{ test k}) \\
&\leq \sum_{i=1}^{k} \Pr(\text{reject } H_0 \text{ test i}) = \alpha_c \times k.
\end{aligned}$$

Solving for α_c we obtain: $\alpha_c = \alpha/k$, which is called Bonferroni correction. By making use of the Boole's inequality, this approach does not require any assumptions about dependence among the tests or about how many of the null hypotheses are true.

Possible solutions

The Bonferroni correction can be conservative if there are a large number of tests, as it comes at the cost of reducing the power of the individual tests (e.g. if $\alpha = 5\%$ and $k = 20$, we get $\alpha_c = 0.05/20 = 0.25\%$). There exists a (slightly) "tighter" bound for α_k, which is given by

$$\alpha_k = \Pr(\text{reject } H_0 \text{ at least once}) \leq 1 - (1 - \alpha_c)^k.$$

Solving for α_c we obtain: $\alpha_c = 1 - (1 - \alpha)^{1/k}$, which is called Dunn–Šidák correction. This correction is (slightly) less stringent than the Bonferroni correction (since $1 - (1 - \alpha)^{1/k} > \alpha/k$ for $k \geq 2$).

There exist many other alternative methods for multiple testing corrections. It is important to mention that when k is large (say > 100) the Bonferroni and Dunn–Šidák corrections become inapplicable and methods based on the idea of False Discovery Rate should be preferred. However, these recent methods are beyond the scope of this class.

Multiple-sample location tests

To compare several means of different populations, a standard approach is to start our analysis by using the multiple-sample location tests. More precisely, we proceed our analysis with the following steps:

- Step 1: We first perform the multiple-sample location tests, where the null hypothesis states that all the locations are the same. If we cannot reject the null hypothesis, we stop our analysis here. Otherwise, we move on to Step 2.
- Step 2: We compare the groups mutually (using α_c) with two-sample location tests in order to verify our hypothesis.

We will discuss three multiple-sample location tests:

1. Fisher's one-way ANalysis Of VAriance (ANOVA)
2. Welch's one-way ANOVA
3. Kruskal-Wallis test

Fisher's one-way ANOVA

This test considers the following assumed model for G groups

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)} = \mu + \delta_g + \varepsilon_{i(g)},$$

where $g = 1, \dots, G$, $i = 1, \dots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2)$ and $\sum n_g \delta_g = 0$.

📝 n_i = sample size of group i, $\mu_i = \mu + \delta_i$ = population mean of group i, $i = 1, \dots, G$.

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_G \quad \text{and} \quad H_a : \mu_i \neq \mu_j \ \text{ for at least one pair of } \ (i, j).$$

Test statistic's distribution under H_0:

$F = \frac{N s_{\bar{X}}^2}{s_p^2} \underset{H_0}{\sim} \text{Fisher}(G-1, N-G)$, where $s_{\bar{X}}^2 = \frac{1}{G-1} \sum_{g=1}^{G} \frac{n_g}{N} (\bar{X}_g - \bar{X})^2$,

$s_p^2 = \frac{1}{N-G} \sum_{g=1}^{G} (n_g - 1) s_g^2$, $N = \sum_{g=1}^{G} n_g$, and $\bar{X} = \frac{1}{N} \sum_{g=1}^{G} n_g \bar{X}_g$.

Discussion - Fisher's one-way ANOVA

Python function:

```
from scipy import stats

stats.f_oneway(group_A, group_B, group_C)
```

This test strongly relies on the assumed absence of outliers. If outliers appear to be present the Kruskal-Wallis test (see later) is (probably) a better option.

For moderate and small sample sizes, the sample distribution should be at least approximately normal with no strong skewness to ensure the reliability of the test.

In practice, the assumption of equal variance is hard to verify so **we recommend to avoid this test in practice**.

Welch's one-way ANOVA

This test considers the following assumed model for G groups

$$X_{i(g)} = \mu_g + \varepsilon_{i(g)} = \mu + \delta_g + \varepsilon_{i(g)},$$

where $g = 1, ..., G$, $i = 1, \ldots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma_g^2)$ and $\sum n_g \delta_g = 0$.

Hypotheses:

$$H_0 : \mu_1 = \mu_2 = ... = \mu_G \quad \text{and} \quad H_a : \mu_i \neq \mu_j \;\; \text{for at least one pair of} \;\; (i, j).$$

Test statistic's distribution under H_0:

$$F^* = \frac{s_{\overline{X}}^{*2}}{1 + \frac{2(G-2)}{3\Delta}} \underset{H_0}{\sim} \text{Fisher}(G-1, \Delta),$$

where $s_{\overline{X}}^{*2} = \frac{1}{G-1} \sum_{g=1}^{G} w_g (\overline{X}_g - \overline{X}^*)^2$, $\Delta = [\frac{3}{G^2-1} \sum_{g=1}^{G} \frac{1}{n_g} (1 - \frac{w_g}{\sum_{g=1}^{G} w_g})]^{-1}$, $w_g = \frac{n_g}{s_g^2}$,

and $\overline{X}^* = \sum_{g=1}^{G} \frac{w_g \overline{X}_g}{\sum_{g=1}^{G} w_g}$.

Discussion - Welch's one-way ANOVA

Python function:

```python
from statsmodels.stats.oneway import anova_oneway

anova_oneway(data, groups=groups, use_var='unequal', welch_correction=True)
```

This test strongly relies on the assumed absence of outliers. If outliers appear to be present the Kruskal-Wallis test (see later) is (probably) a better option.

For moderate and small sample sizes, the sample distribution should be at least approximately normal with no strong skewness to ensure the reliability of the test.

This test does not require the variances of the groups to be equal. If the variances of all the groups are the same (which is rather unlikely in practice), the Welch's one-way ANOVA loses a little bit of power compared to the Fisher's one-way ANOVA.

Kruskal-Wallis test

This test considers the following assumed model for G groups

$$X_{i(g)} = \theta_g + \varepsilon_{i(g)} = \theta + \delta_g + \varepsilon_{i(g)},$$

where $g = 1, \dots, G$, $i = 1, \dots, n_g$, $\varepsilon_{i(g)} \overset{iid}{\sim} N(0, \sigma^2)$ and $\sum n_g \delta_g = 0$.

📝 n_i = sample size of group i, $\theta_i = \theta + \delta_i$ = population location of group i, $i = 1, \dots, G$.

Hypotheses:

$$H_0 : \theta_1 = \theta_2 = \dots = \theta_G \quad \text{and} \quad H_a : \theta_i \neq \theta_j \quad \text{for at least one pair of} \quad (i, j).$$

Test statistic's distribution under H_0 : $\quad H = \dfrac{\frac{12}{N(N+1)} \sum_{g=1}^{G} \frac{\overline{R}_g}{n_g} - 3(N-1)}{1 - \frac{\sum_{v=1}^{V} t_v^3 - t_v}{N^3 - N}} \underset{H_0}{\sim} \mathcal{X}(G-1)$

where $\overline{R}_g = \frac{1}{n_g} \sum_{i=1}^{n_g} R_{i(g)}$ with $R_{i(g)}$ denoting the global rank of the i^{th} observation of group g,

V is the number of different values/levels in X and t_v denotes the number of times a given

value/level occurred in X.

This document was prepared with the help of Lionel Voirol and Filippo Salmaso

Discussion - Kruskal-Wallis test

Python function:

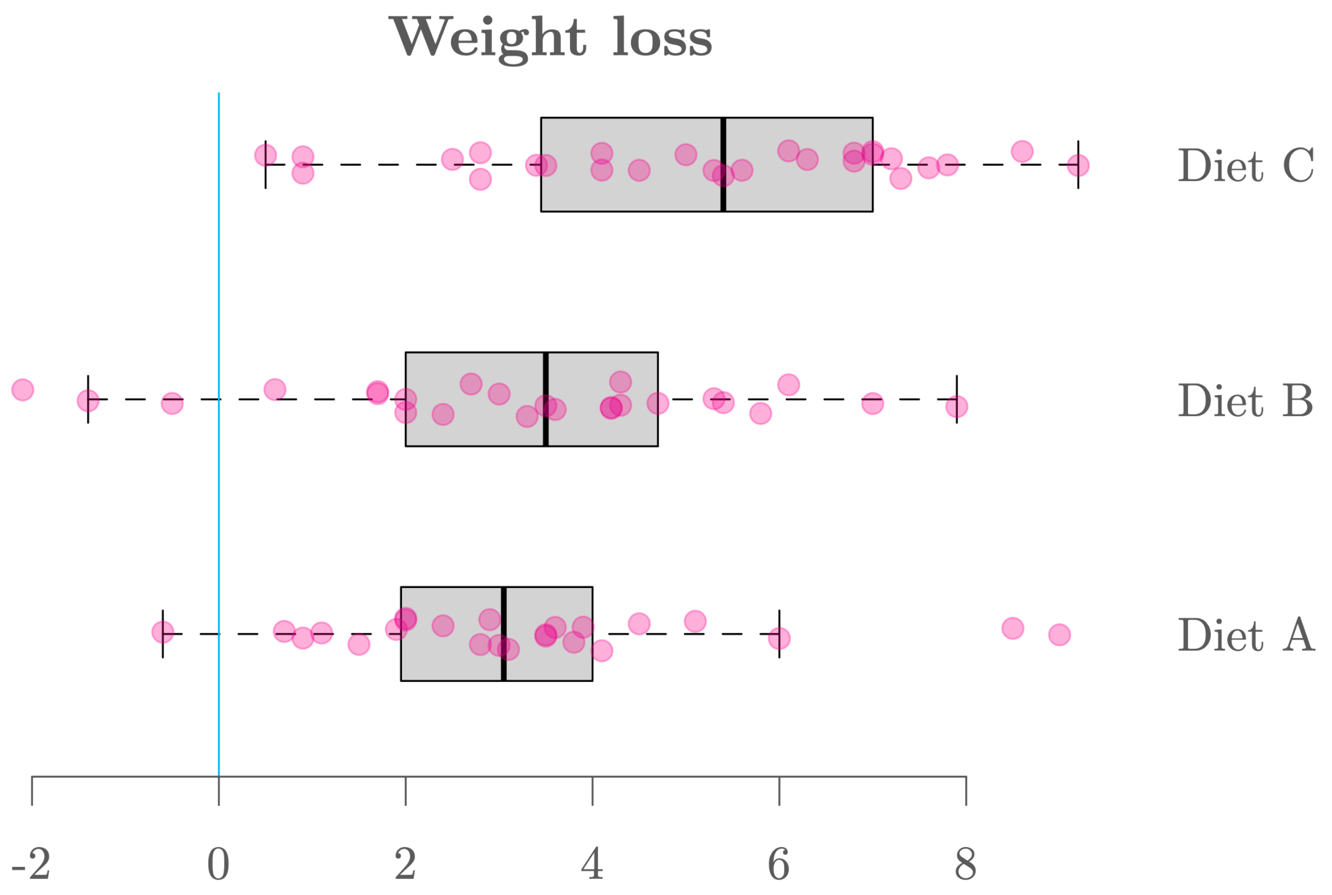
```
from scipy import stats

stats.kruskal(group_A, group_B, group_C)
```

- This test is "robust" in the sense that (unlike the one-way ANOVA) it is not overly affected by outliers.
- For the Kruskal-Wallis test to be comparable to the one-way ANOVAs (i.e. testing for the mean) we need to assume: (1) The distributions are symmetric, (2) the variances are the same. Then, we have $\theta_i = \mu_i, i = 1, \ldots, G$.
- Compared to the one-way ANOVAs, the Kruskal-Wallis test is less powerful if their requirements (Gaussian and possibly same variances) are met.

Exercise: Comparing diets A, B and C

Graph



Exercise: Comparing diets A, B and C

Import

```python
# Import data
import pandas as pd
diet = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Data/
diet["weight.loss"] = diet["initial.weight"] - diet["final.weight"]

# Variable of interest
dietA = diet["weight.loss"][diet["diet.type"]=="A"]
dietB = diet["weight.loss"][diet["diet.type"]=="B"]
dietC = diet["weight.loss"][diet["diet.type"]=="C"]

# Create data frame
dat = pd.DataFrame({
    "response": list(dietA) + list(dietB) + list(dietC),
    "groups": (["A"] * len(dietA)) + (["B"] * len(dietB)) + (["C"] * len(dietC))
})
```

Motivating Example: Reading Ability

Problem

An educator believes that new directed reading activities in the classroom can help elementary school students (6-12 years old) improve their reading ability. She arranged a pilot study where some students (chosen at random) of age 6 start to take part in these activities **(treatment group)**, meanwhile other students continue with the classical curriculum **(control group)**. The educator wishes to evaluate the effectiveness of these activities so all students are given a Degree of Reading Power (DRP) test, which assesses their reading ability.

Can we conclude that these new directed reading activities can help elementary school students improve their reading ability?

Motivating Example: Reading Ability

Data

```python
import pandas as pd

reading = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Da

control = reading.loc[reading["group"] == "Control", "score"]
print(control.head())
```

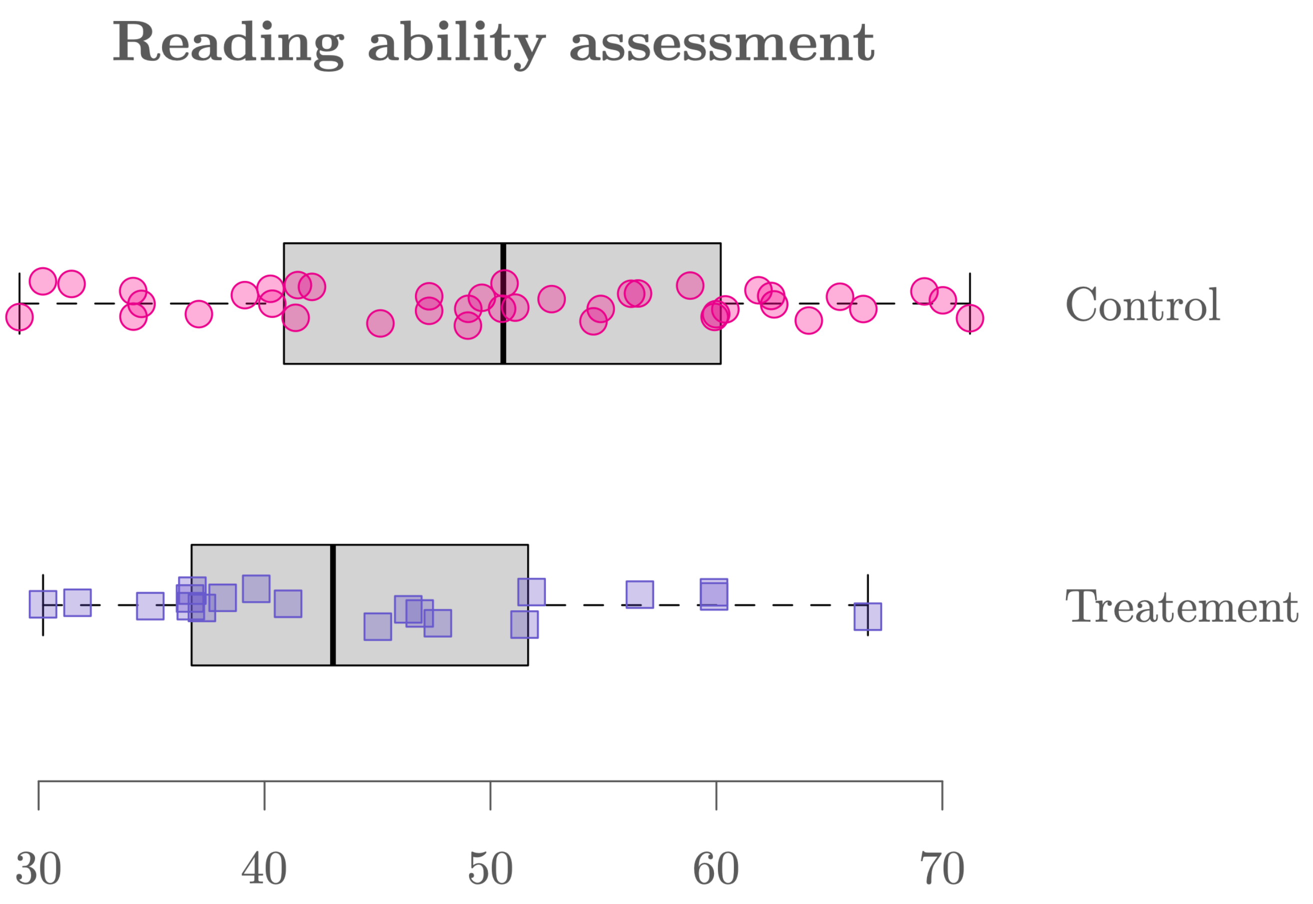
```
0    59.988809
1    60.394867
2    70.023947
3    41.477072
4    29.154968
Name: score, dtype: float64
```

```python
treatment = reading.loc[reading["group"] == "Treatment", "score"]
print(treatment.head())
```

```
40    36.805579
41    36.702008
42    59.893620
43    31.722885
44    41.042415
Name: score, dtype: float64
```

Motivating Example: Reading Ability

Graph



Motivating Example: Reading Ability

Test

1. Define hypotheses: $H_0 : \mu_T = \mu_C$ and $H_a : \mu_T > \mu_C$.
2. Define α: We consider $\alpha = 5\%$.
3. Compute p-value: p-value = 97.12% (see Python code tab for details).
4. Conclusion: We have p-value > α so we cannot reject the null hypothesis at the significance level of 5%.

Remark: Notice that the p-value for the test with opposite hypotheses is actually $100\% - 97.12\% = 2.88\% < \alpha$ (Why? 🤔). So we conclude, at the significance level of 5%, that the classical curriculum without these new directed reading activities actually improve students' reading ability more compared to the curriculum with these activities. 😬

Motivating Example: Reading Ability

Python Code

```python
t_test_reading, p_value_reading = stats.ttest_ind(
    treatment, control, equal_var=True, alternative='greater'
)

print(f"t-test: {t_test_reading:.4f}")
```

```
t-test: -1.8553
```

```python
print(f"p-value: {p_value_reading:.4f}")
```

```
p-value: 0.9657
```

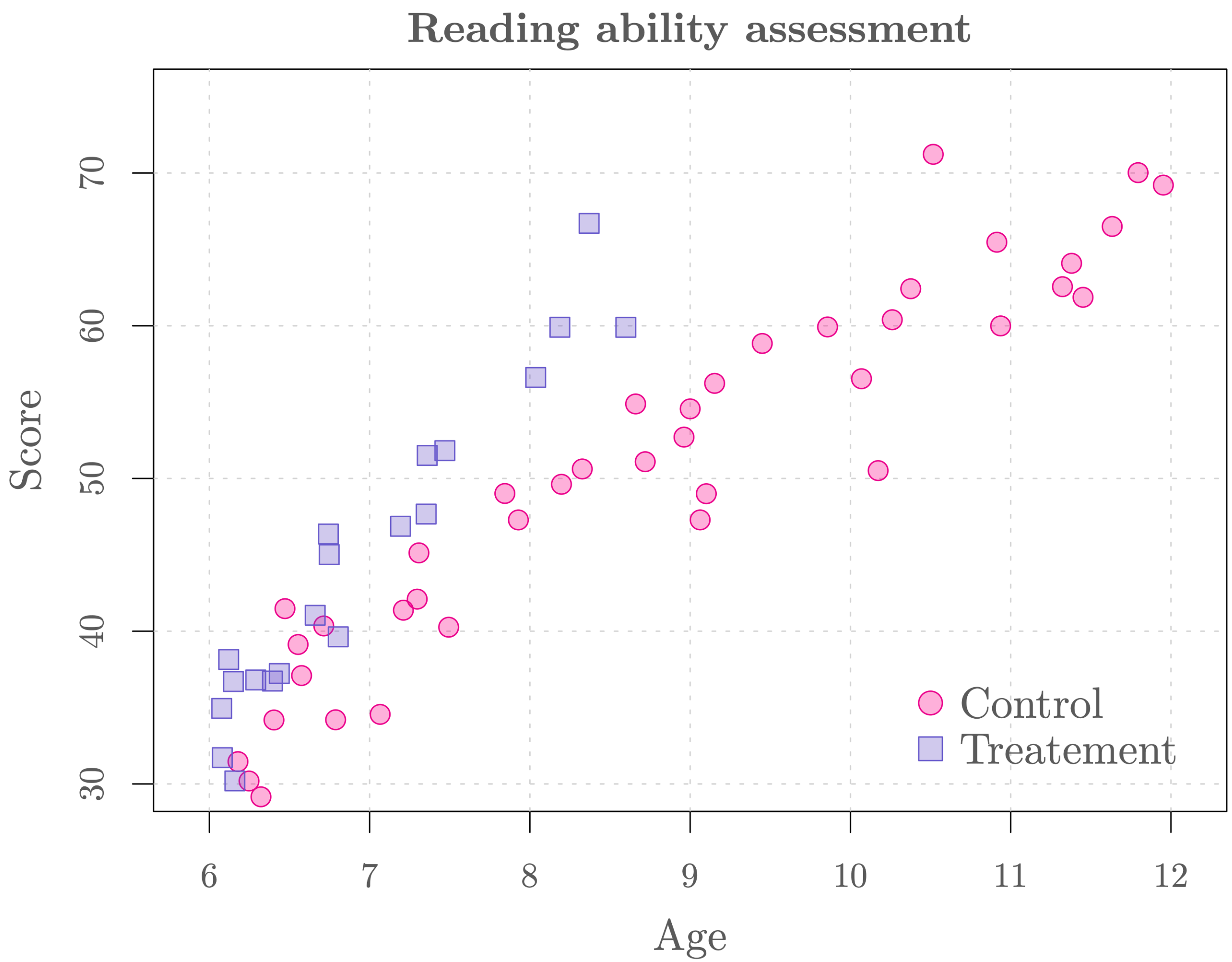
Is our analysis comprehensive?

The educator points out that only students of 6-8 years old have participated in the new directed reading activities. In other words, in the sample she collected, the students in the treatment group are only of age 6 to 8, whereas the students in the control group vary from 6 to 12 years old. Is age a potential explanation to the difference we observe among the students' reading ability?

To make sure that the analysis is reliable, she includes the age information of the students, which can be accessed as follows:

```
treatment_age = reading.loc[reading["group"] == "Treatment", "age"]
control_age   = reading.loc[reading["group"] == "Control", "age"]
```

Should age be taken into account?



Regression analysis

- Regression analysis corresponds to a set of statistical methods for estimating the relationships between a response variable Y of primary interest (also called the *outcome variable*) and some explanatory variables $X_1, \dots, X_p$ (also called *covariates*, *regressors*, *features* or *predictors*).
- The relationship between the response variable Y and the covariates is not deterministic and we model the conditional expected value (i.e. $\mathbb{E}[Y|X_1, \dots, X_p]$).
- Therefore, we consider the following (general) model:

$$Y_i = \mathbb{E}[Y_i|X_{i1}, \dots, X_{ip}] + \varepsilon_i,$$

  where $\mathbb{E}[\varepsilon_i] = 0$ and $i = 1, \dots, n$.
- Example: $\mathbb{E}[\text{reading abilities}_i | \text{ age}_i, \text{treatment}_i, \dots]$.

Linear regression

- The most common form of regression analysis is linear regression, in which the conditional expected value $\mathbb{E}[Y|X_1, \ldots, X_p]$ takes the form

$$\mathbb{E}[Y_i|X_{i1}, \ldots, X_{ip}] = \beta_0 + \beta_1 X_1 + \ldots + \beta_p X_p.$$

and $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$.

- Our general model can be expressed as

$$Y_i = \mathbb{E}[Y_i|X_{i1}, \ldots, X_{ip}] + \varepsilon_i = \beta_0 + \sum_{j=1}^{p} \beta_j X_{ij} + \varepsilon_i,$$

and therefore,

$$Y_i \sim N\left(\beta_0 + \sum_{j=1}^{p} \beta_j X_{ij}, \sigma^2\right).$$

Linear regression

Therefore, this approach makes two (strong) assumptions:

1. The conditional expected value $\mathbb{E}[Y|X_1, \dots, X_p]$ is assumed to be a linear function of the covariates.
2. The errors are assumed to be *iid* Gaussian, i.e. $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, at least when the sample size is small to medium.

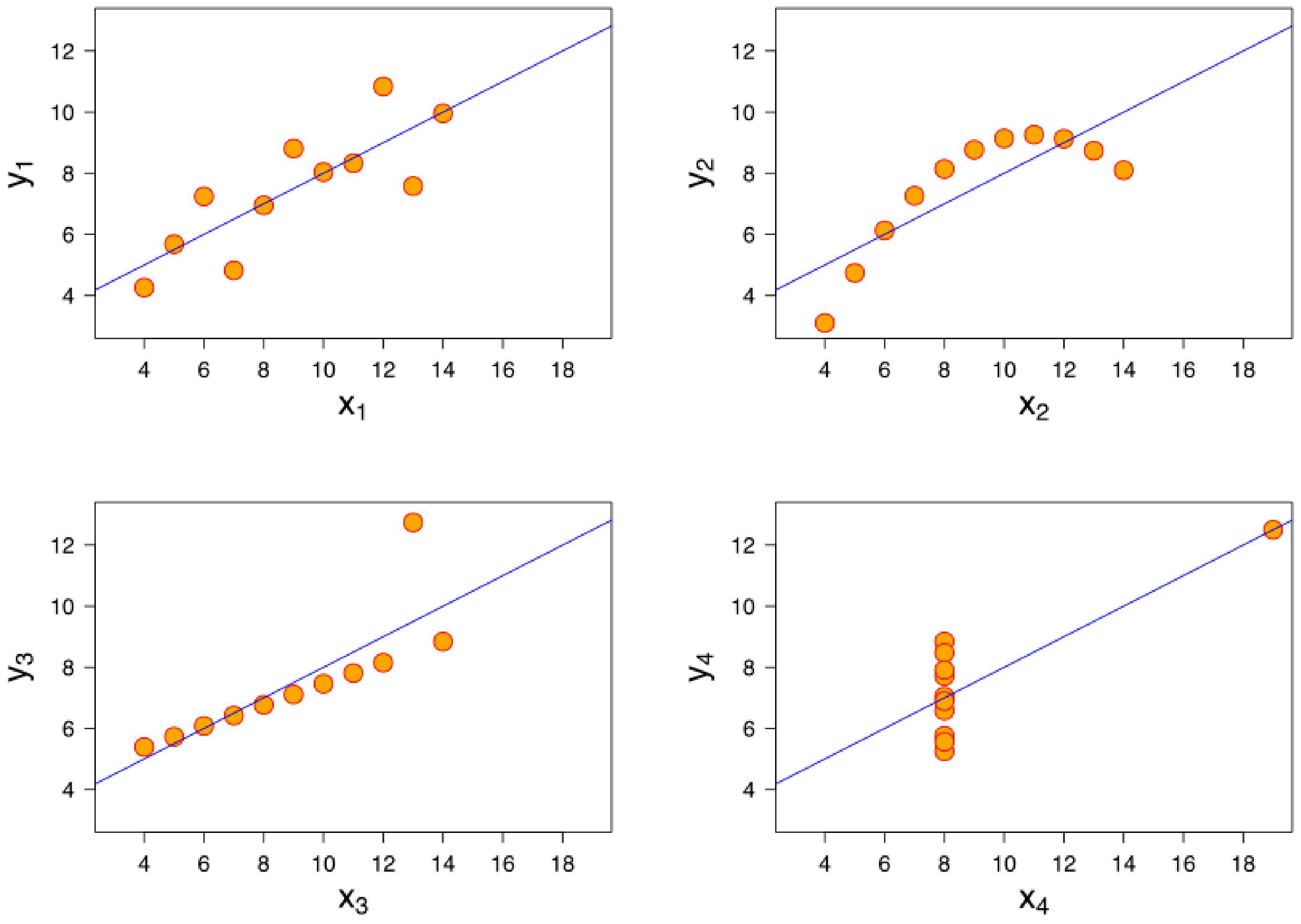
⚠️ In practice, it is important to assess if these assumptions are plausible.

The parameters of the model (i.e. $\beta_0, \beta_1, \dots, \beta_p$ and σ^2) can be estimated by several methods. The most commonly used is the least squares approach where $\beta_0, \beta_1, \dots, \beta_p$ are chosen such that

$$\sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} \left(Y_i - \mathbb{E}[Y_i|X_{i1}, \dots, X_{ip}]\right)^2 = \sum_{i=1}^{n} \left(Y_i - \beta_0 - \sum_{j=1}^{p} \beta_j X_{ij}\right)^2$$

is minimized, which then allows to estimate σ^2 further on.

Anscombe's quartet



👋 Source: Wikipedia.

Example: Reading ability assessment

In the reading ability example, we can formulate a linear regression model (without interaction) as follows:

$$\text{Score}_i = \beta_0 + \beta_1 \text{Group}_i + \beta_2 \text{Age}_i + \epsilon_i, \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

- Score_i: score of the DRP test of the i-th student.
- Group_i: indicator of participation of the new directed reading activities for the i-th student (i.e. $\text{Group}_i = 1$ if participate and $\text{Group}_i = 0$ if not participate).
- Age_i: age of the i-th student (related to *time since start of treatment*).

With this model the two groups can be compared as the age effect is taken into account. The goal of the educator is now to assess if β_1 is significantly larger than 0.

Example: Reading ability assessment

`Python` *Code*

Python function:

```python
import statsmodels.formula.api as smf

# Fit linear model using formula syntax
model = smf.ols(formula="y ~ x1 + x2 + ... + xp", data=mydata).fit()

# Show results
print(model.summary())
```

Example: Reading ability assessment

Output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  score   R-squared:                       0.859
Model:                            OLS   Adj. R-squared:                  0.854
Method:                 Least Squares   F-statistic:                     173.0
Date:                Wed, 22 Oct 2025   Prob (F-statistic):           6.16e-25
Time:                        14:04:21   Log-Likelihood:                -173.13
No. Observations:                  60   AIC:                             352.3
Df Residuals:                      57   BIC:                             358.6
Df Model:                           2
Covariance Type:            nonrobust
======================================================================================
                         coef    std err          t      P>|t|      [0.025      0.975]
--------------------------------------------------------------------------------------
Intercept             -7.8639      3.324     -2.366      0.021     -14.520      -1.208
group[T.Treatment]     6.3771      1.393      4.578      0.000       3.587       9.167
age                    6.6457      0.370     17.985      0.000       5.906       7.386
==============================================================================
Omnibus:                        0.633   Durbin-Watson:                   2.068
Prob(Omnibus):                  0.729   Jarque-Bera (JB):                0.386
Skew:                           0.196   Prob(JB):                        0.824
Kurtosis:                       3.013   Cond. No.                         50.7
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Interpretation of coefficients

We can obtain the estimated coefficients. Specifically,

- $\hat{\beta}_0 = -7.8639$ represents the estimated baseline average score of the DRP test at birth (but what does it mean? 🤨).
- $\hat{\beta}_1 = 6.3771$ means that for a student of the same age, participating in the new directed reading activities is estimated to increase their average score of the DRP test by 6.3771.
- $\hat{\beta}_2 = 6.6457$ means that when a student receives the same treatment (either participate or not in the activities), their average score increases by 6.6457 as they become 1 year older.

Regression coefficients represent the mean change in the response variable for one unit of change in the predictor variable while holding other covariates in the model constant.

Interpretation of coefficient p-values

- We notice that for each coefficient β_j, there is a corresponding p-value associated to the (Wald t-)test of $H_0 : \beta_j = 0$ and $H_a : \beta_j \neq 0$.
- A covariate with a small p-value (typically smaller than 5%) is considered to be a significant (meaningful) addition to the model, as changes in the values of such covariate can lead to changes in the response variable.
- On the other hand, a large p-value (typically larger than 5%) suggests that the corresponding covariate is not (significantly) associated with changes in the response or that we don't have enough evidence (data) to show its effect.
- In this example, the coefficient p-value associated to the `group` covariate is 2.6×10^{-3}%. This suggests that taking into account the effect of `age`, the reading abilities of the students receiving the treatment is significantly **different** from the control group, at the significance level of 5%. But this is not what we want!

Interpretation of coefficient p-values

In the linear regression output, the coefficient p-value (which we denote as p below) corresponds to a two-sided test. We can use this result to compute the p-value of a one-sided test using the following relations:

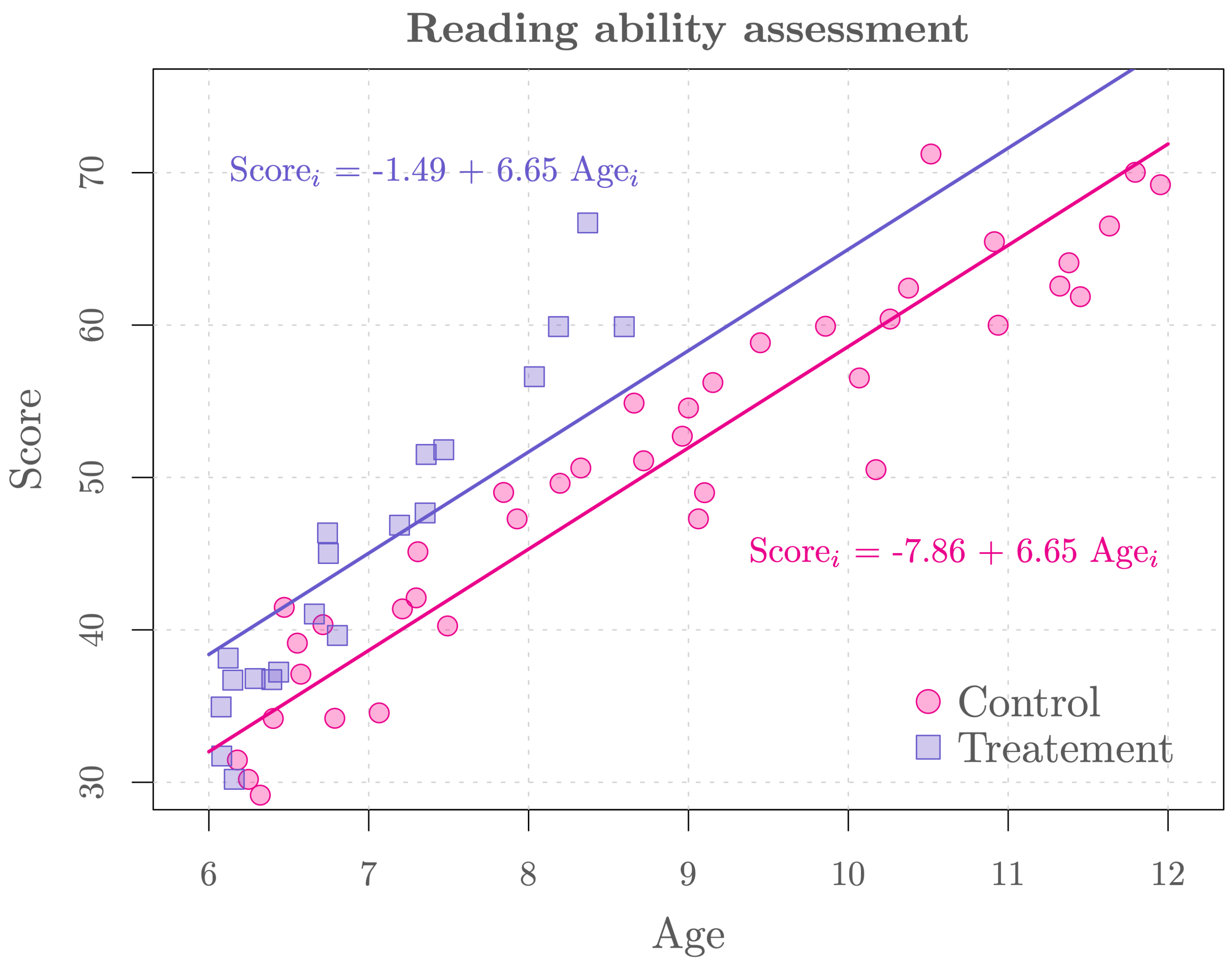
	$H_a : \beta_j > 0$	$H_a : \beta_j < 0$
$\hat{\beta}_j > 0$	✅ Correct tail (prob = $p/2$)	⚠️ Wrong tail (prob = $1 - p/2$)
$\hat{\beta}_j < 0$	⚠️ Wrong tail (prob = $1 - p/2$)	✅ Correct tail (prob = $p/2$)

✅ = rejection region corresponds to the alternative
⚠️ = rejection region is in the opposite direction of the alternative

In our example, $\beta_1 = 6.3771$ and $p = 2.6 \times 10^{-3}\%$. So the p-value of the test with hypotheses $H_0 : \beta_1 = 0$ and $H_a : \beta_1 > 0$ is $2.6 \times 10^{-3}\%/2 \approx 1.3 \times 10^{-3}\% < \alpha$. So we can conclude that these new directed reading activities can significantly improve students' reading ability compared to classical curriculum.

However, is our model plausible? 🤔

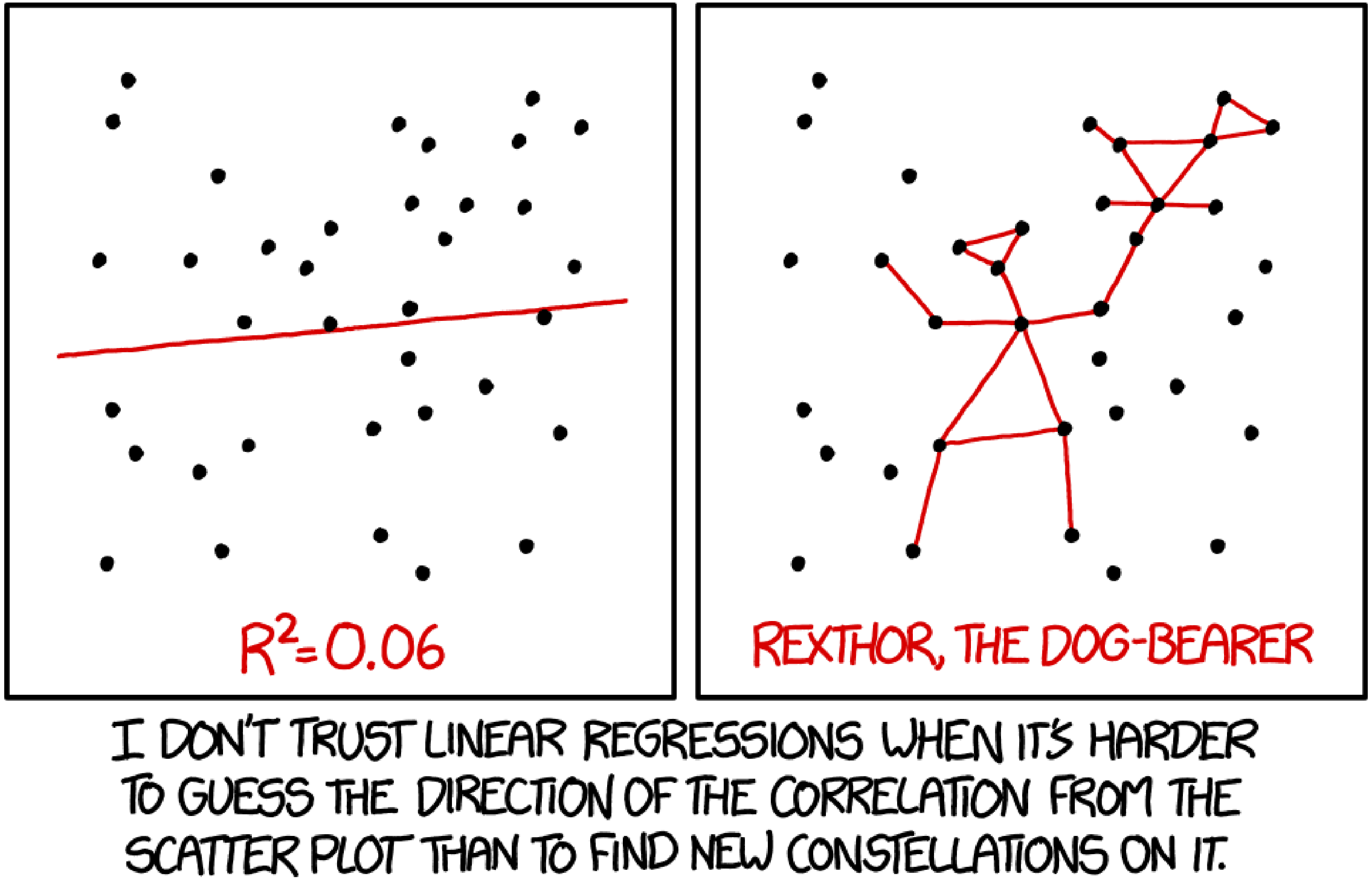
How good is our model? 🤔



Could we use the R^2?

- The coefficient of determination, denoted as R^2 and often referred to as R-squared, corresponds to the proportion of the variance in the response variable that is “explained” by the model.
- R^2 gives certain information about the goodness of fit of a model. It measures how well the regression predictions approximate the real data points. An R^2 of 1 indicates that the regression predictions perfectly fit the data.
- However, the value of R^2 is not related to the adequacy of our model to the data.
- ⚠️ Moreover, adding new covariates to the current model always increases R^2, whether the additional covariates are significant or not. Therefore, R^2 alone cannot be used as a meaningful comparison of models with different covariates.
- The adjusted R^2 is a modification of R^2 that aims to limit this issue.

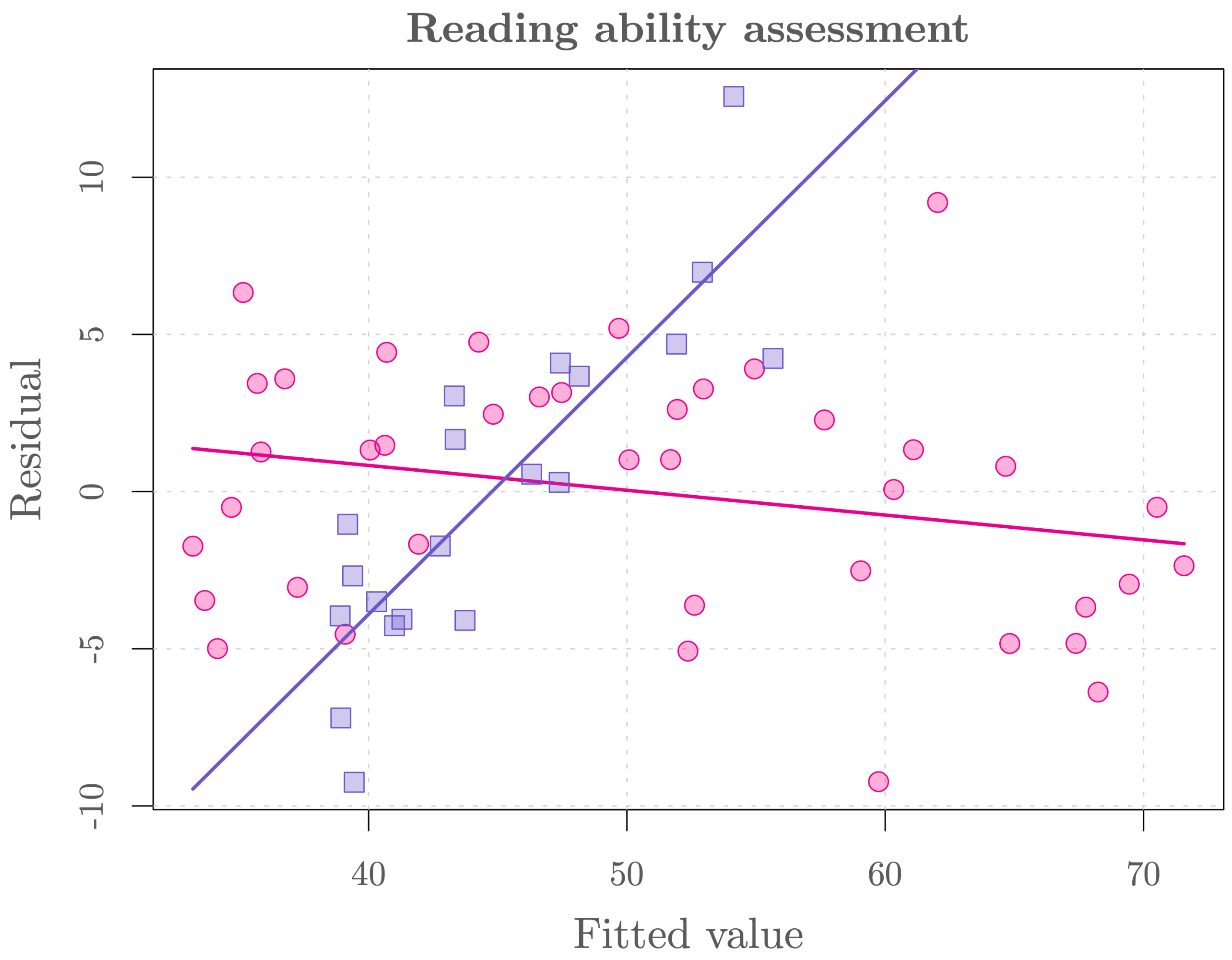
Rexthor, the Dog-Bearer!



👋 If you want to know more have a look here.

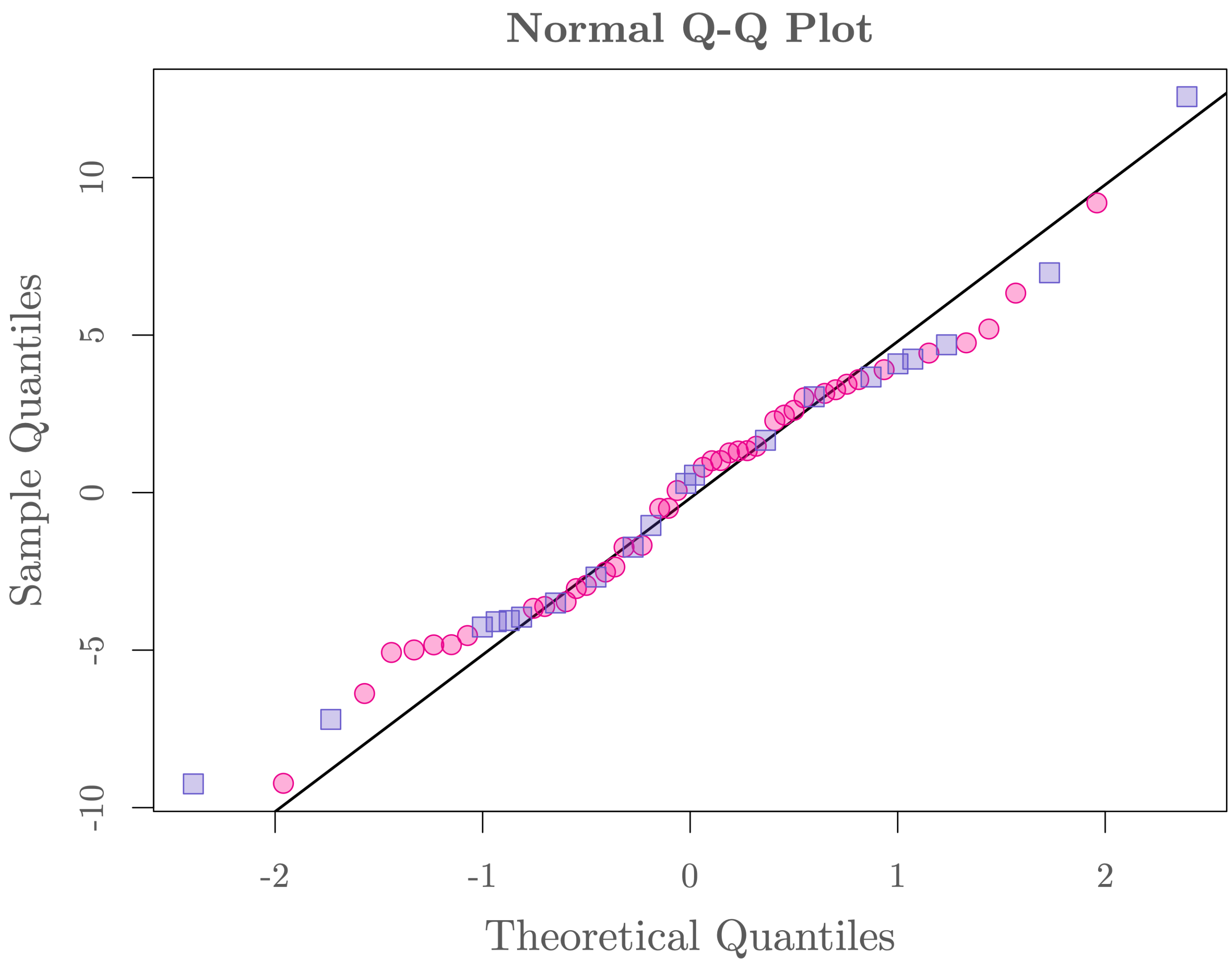
Model diagnostic

Model diagnostic ⚠️



Model diagnostic ⚠️

Model diagnostic ⚠️



Let's update our model

Our results suggest that the students of the group participating in these new directed reading activities progress more rapidly (which is actually more reasonable than our initial model 🤔). Therefore, we modify our model by adding an interaction term:

$$\text{Score}_i = \beta_0 + \beta_1 \text{Group}_i + \beta_2 \text{Age}_i + \beta_3 \text{Group}_i \text{Age}_i + \epsilon_i, \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

- Score_i: score of the DRP test of the i-th student.
- Group_i: indicator of participation of the new directed reading activities for the i-th student (i.e. $\text{Group}_i = 1$ if participate and $\text{Group}_i = 0$ if not participate).
- Age_i: age of the i-th student (related to *time since start of treatment*),

The goal of the educator is now to assess if β_1 and/or β_3 are significantly larger than 0.

Example: Reading ability assessment

Python Code

Here is the code to fit our second model:

```
import pandas as pd
import statsmodels.formula.api as smf

# Import data (if you haven't already)
reading = pd.read_csv("/Users/filipposalmaso/Documents/Temporary_Stuff/Data-Science_copy/r

# Fit a linear regression model
mod2 = smf.ols(formula="score ~ group*age", data=reading).fit()

# Show summary
print(mod2.summary())
```

Example: Reading ability assessment

Output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  score   R-squared:                       0.910
Model:                            OLS   Adj. R-squared:                  0.906
Method:                 Least Squares   F-statistic:                     189.6
Date:                Wed, 22 Oct 2025   Prob (F-statistic):           2.70e-29
Time:                        14:04:22   Log-Likelihood:                -159.45
No. Observations:                  60   AIC:                             326.9
Df Residuals:                      56   BIC:                             335.3
Df Model:                           3
Covariance Type:            nonrobust
==========================================================================================
                             coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------------
Intercept                 -3.2489      2.790     -1.164      0.249      -8.839       2.341
group[T.Treatment]       -36.0307      7.539     -4.779      0.000     -51.134     -20.928
age                        6.1207      0.311     19.693      0.000       5.498       6.743
group[T.Treatment]:age     5.9539      1.047      5.688      0.000       3.857       8.051
==============================================================================
Omnibus:                        0.210   Durbin-Watson:                   2.013
Prob(Omnibus):                  0.901   Jarque-Bera (JB):                0.005
Skew:                           0.013   Prob(JB):                        0.998
Kurtosis:                       3.036   Cond. No.                         145.
==============================================================================

Notes:
```

Interpretation of coefficients

We can obtain the estimated coefficients. Specifically,

- $\hat{\beta}_0 = -3.2489$ represents the estimated baseline average score of the DRP test at birth (but *again* what does it mean? 🙄)
- $\hat{\beta}_1 = -36.0307$ means that for a student of the same age, participating in the new directed reading activities is estimated to decrease their average score of the DRP test by 36.0307 (does this make sense? 🧐).
- $\hat{\beta}_2 = 6.1207$ means that for students not participating to the new directed reading activities, their average score increases by 6.1207 as they become 1 year older.
- $\hat{\beta}_3 = 5.9539$ means that the average score of students participating in the new directed reading activities increases by 5.9539 as they become 1 year older compared to the other students. This means that the average score of students participating to the new program increases by 12.0746 (i.e. 6.1207 + 5.9539) as they become 1 year older.

Model fit

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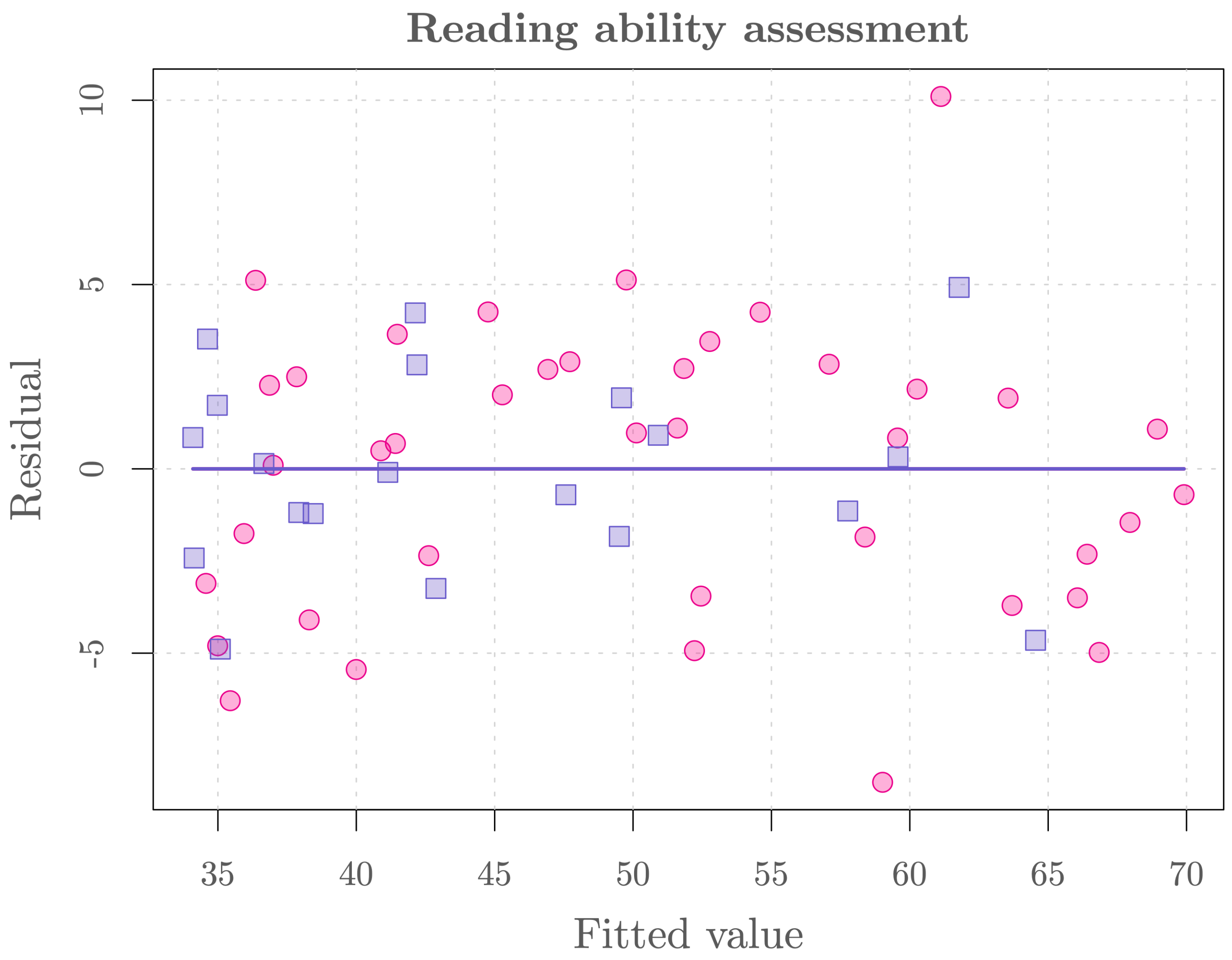
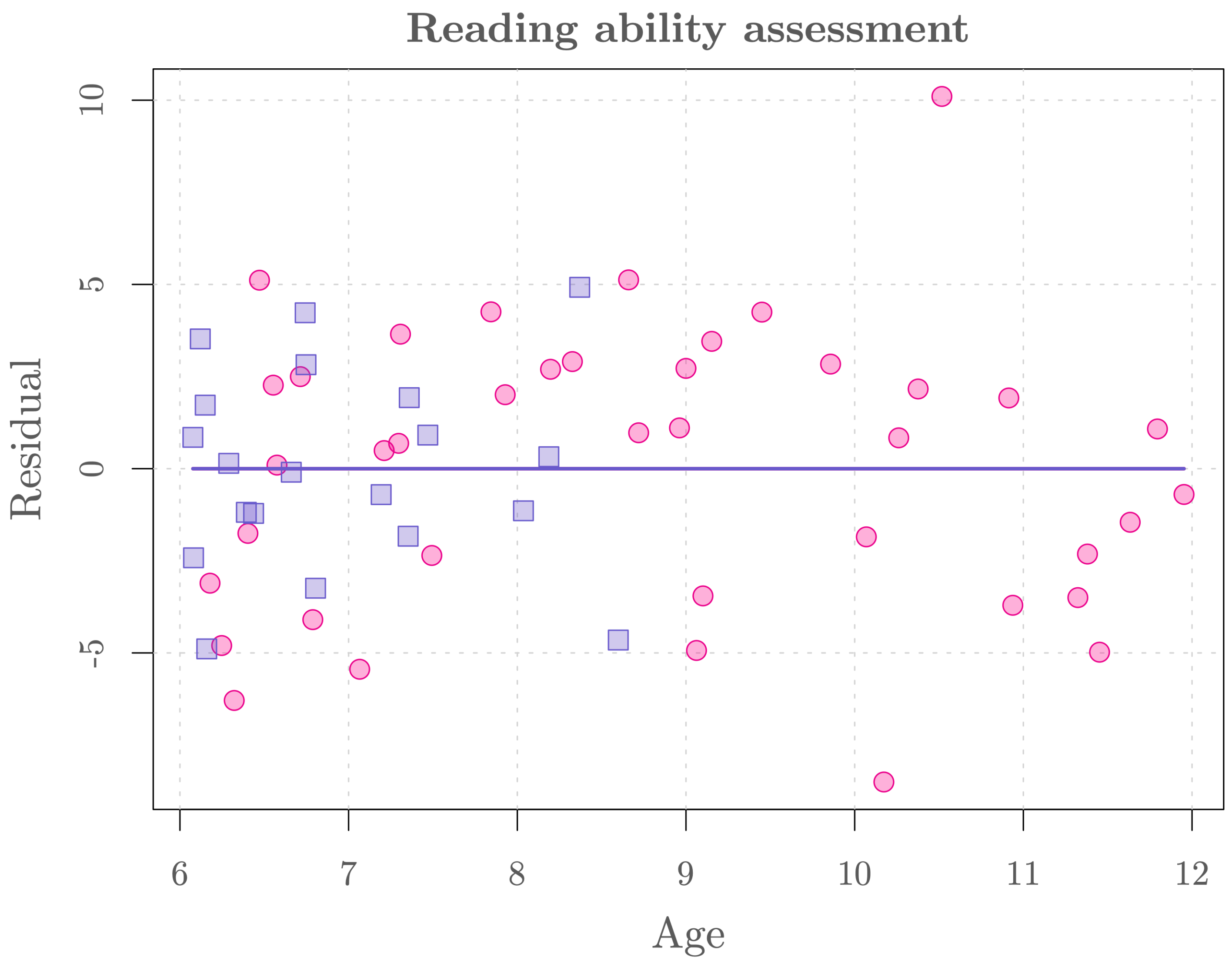
Model diagnostic



This document was prepared with the help of Lionel Voirol and Filippo Salmaso

Model diagnostic



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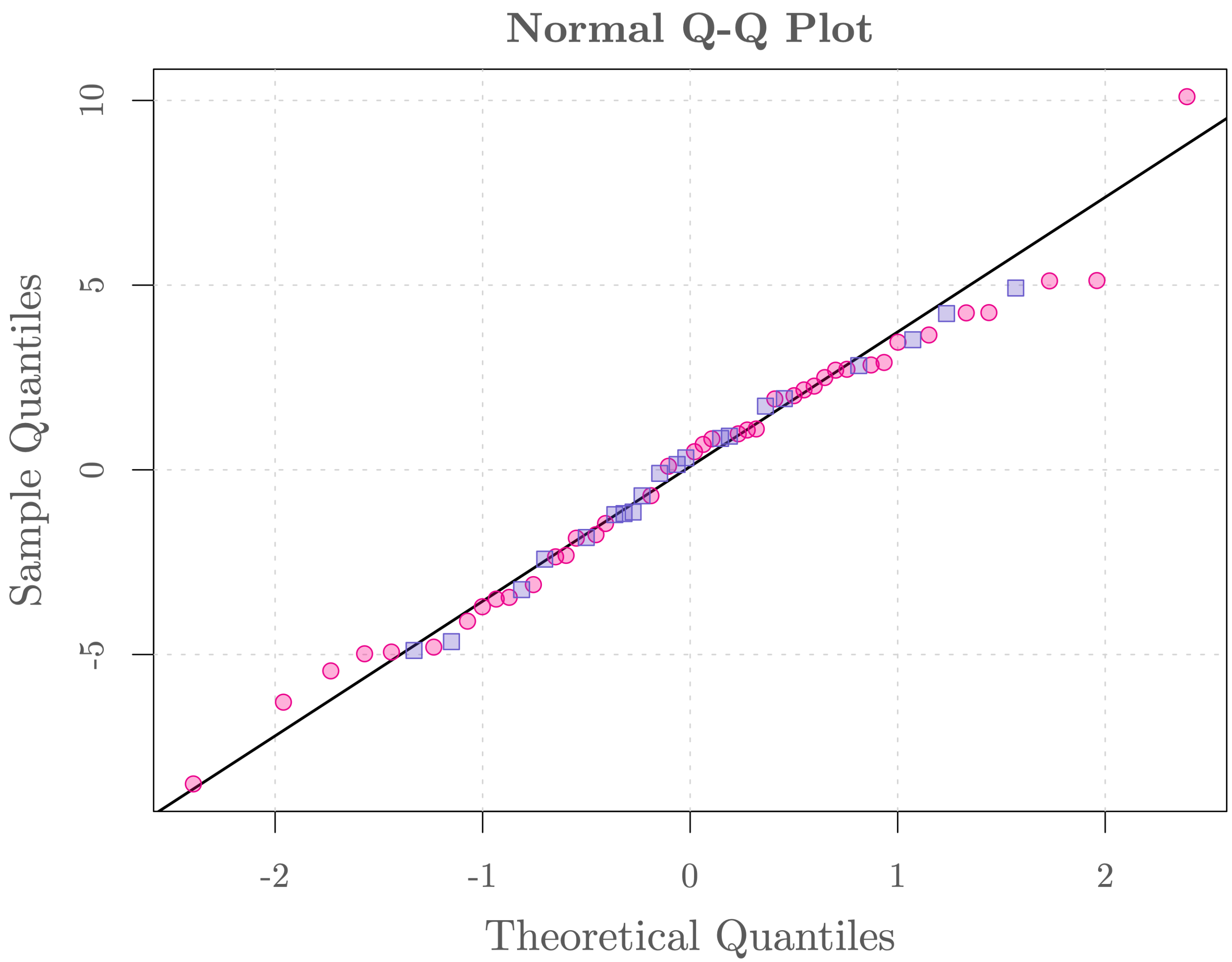
Model diagnostic



Model selection

In general, we prefer a parsimonious approach to modeling in the sense that we only choose a more complex model if the benefits are "substantial"👋. We want our model to be such that:

1. The model fits the data well.
2. Avoid (excessive) overfitting.

Naturally these two objectives are contradictory and there are many ways to select a suitable model (actually this is one of the most active areas of research in modern Statistics). In this class, we will only consider one (simple) approach based on the Akaike Information Criterion (AIC). This criterion corresponds to an estimator of prediction error and thereby relative quality of statistical models for a given set of data.

👋 This point of view is based on Occam's razor (or law of parsimony), the problem-solving principle stipulating that "the simplest explanation is usually the right one".

Model selection

In `Python`, the AIC can be computed for a given model (i.e. use the output of the function `sfm.ols(...)` in `model.aic`.) For example, we can compare the AIC of the previously considered models as follows:

```
print(round(mod1.aic, 4))      # First model (no interaction)
```

```
352.2688
```

```
print(round(mod2.aic, 4))      # Second model (with interaction)
```

```
326.9095
```

As expected, these results suggest that the second model is more appropriate. But should we further improve it?

Let's update our model (again)

It should be reasonable that the average reading scores of the two groups are the same at the start of the program.

$$\text{Score}_i = \beta_0 + \beta_1(\text{Age}_i - 6) + \beta_2\text{Group}_i(\text{Age}_i - 6) + \epsilon_i, \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2).$$

- Score_i: score of the DRP test of the i-th student.
- Group_i: indicator of participation of the new directed reading activities for the i-th student (i.e. $\text{Group}_i = 1$ if participate and $\text{Group}_i = 0$ if not participate).
- $\text{Age}_i - 6$: corresponds to the time since start of treatment of the i-th student.

With this model the two groups can be compared as the age effect is taken into account. The goal of the educator now is (only!) to assess if β_1 is significantly larger than 0.

Example: Reading ability assessment

Python Code

Here is the code to fit our third model:

```
import pandas as pd
import statsmodels.formula.api as smf

# Import data (if you haven't already)
reading = pd.read_csv("https://raw.githubusercontent.com/ELSTE-Master/Data-Science/main/Da
reading["age_minus_six"] = reading["age"] - 6

# Fit a linear regression model
mod3 = smf.ols(formula="score ~ age_minus_six + group:age_minus_six", data=reading).fit()

# Show summary
print(mod3.summary())
```

Example: Reading ability assessment

Output

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  score   R-squared:                       0.910
Model:                            OLS   Adj. R-squared:                  0.907
Method:                 Least Squares   F-statistic:                     289.2
Date:                Wed, 22 Oct 2025   Prob (F-statistic):           1.42e-30
Time:                        14:04:22   Log-Likelihood:                -159.47
No. Observations:                  60   AIC:                             324.9
Df Residuals:                      57   BIC:                             331.2
Df Model:                           2
Covariance Type:            nonrobust
====================================================================================================
                                       coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------------------
Intercept                           33.3504      0.790     42.192      0.000      31.768      34.933
age_minus_six                        6.1522      0.260     23.625      0.000       5.631       6.674
group[T.Treatment]:age_minus_six     5.8103      0.716      8.119      0.000       4.377       7.243
==============================================================================
Omnibus:                        0.192   Durbin-Watson:                   2.013
Prob(Omnibus):                  0.909   Jarque-Bera (JB):                0.008
Skew:                           0.027   Prob(JB):                        0.996
Kurtosis:                       3.020   Cond. No.                         5.84
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Example: Reading ability assessment

AIC

```r
print(round(mod1.aic, 4))
```

```
352.2688
```

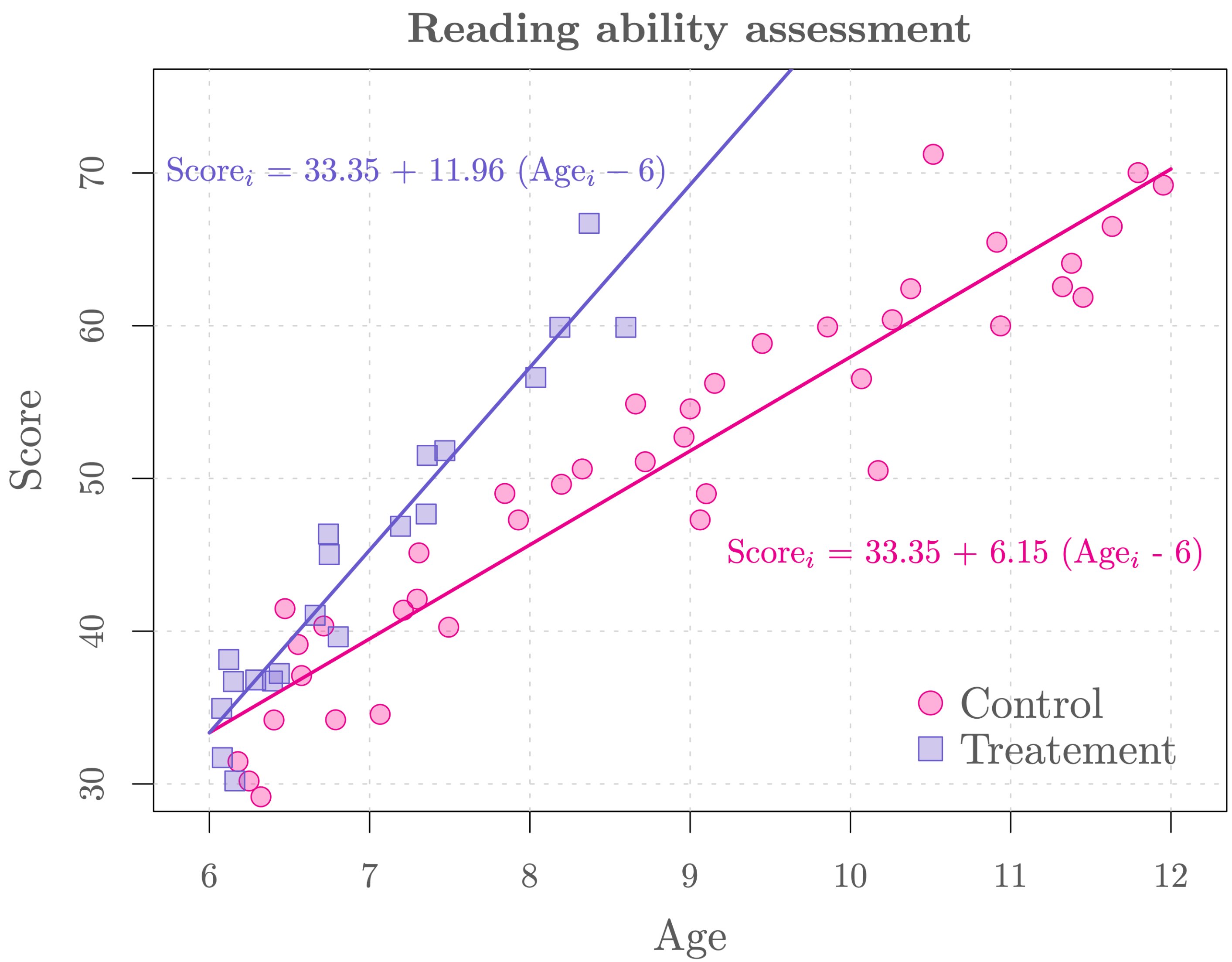
```r
print(round(mod2.aic, 4))
```

```
326.9095
```

```r
print(round(mod3.aic, 4))
```

```
324.9479
```

Model fit



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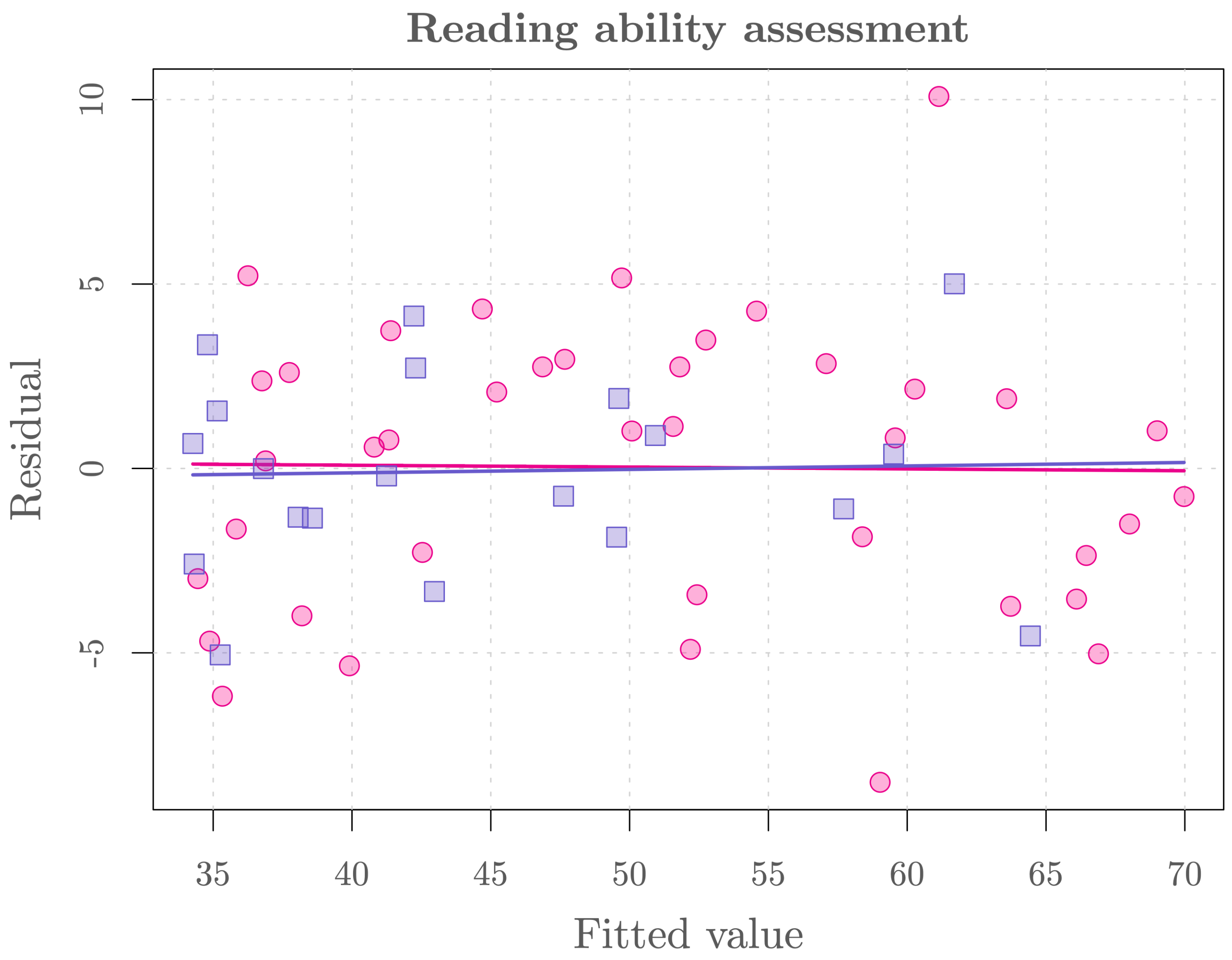
Model diagnostic



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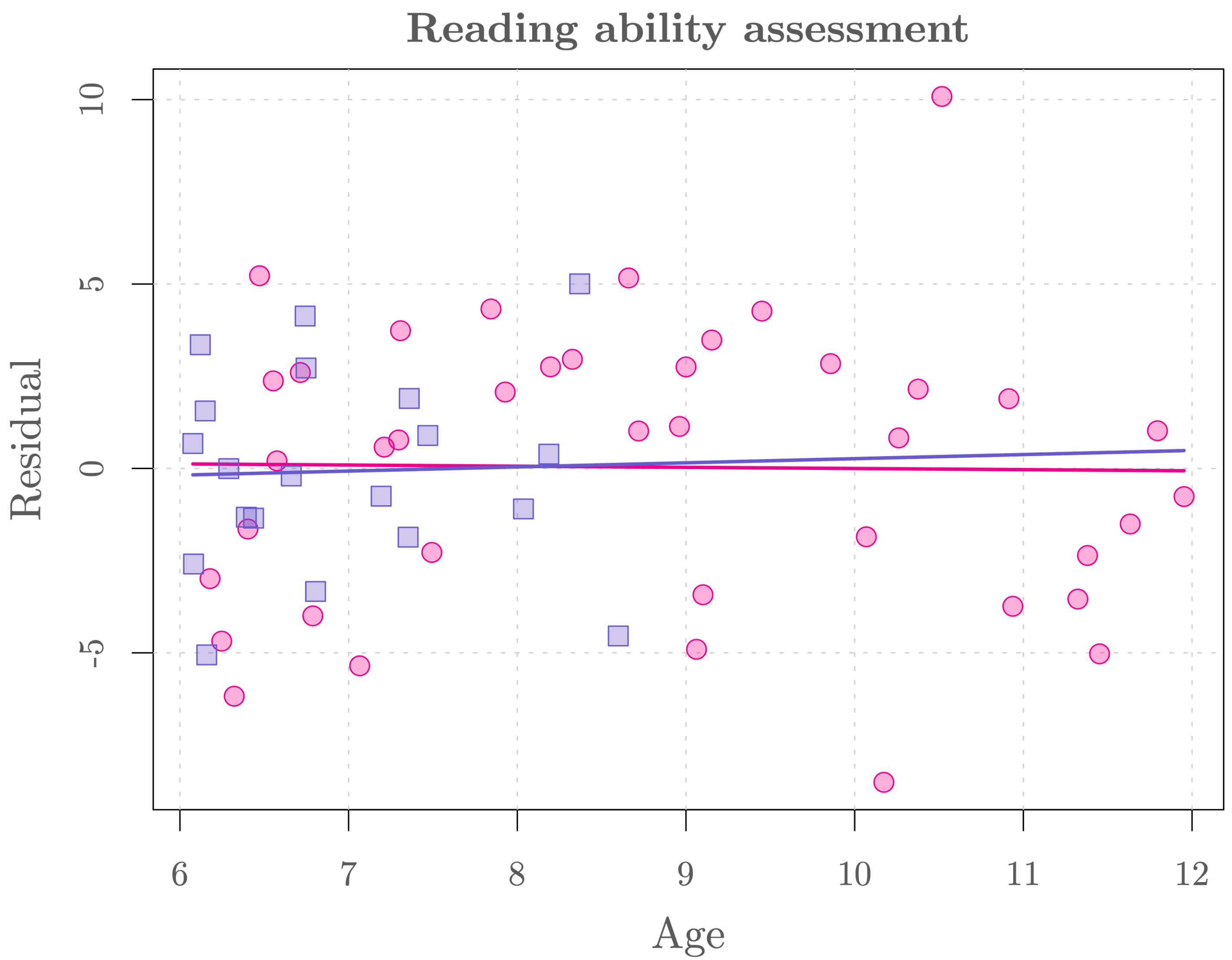

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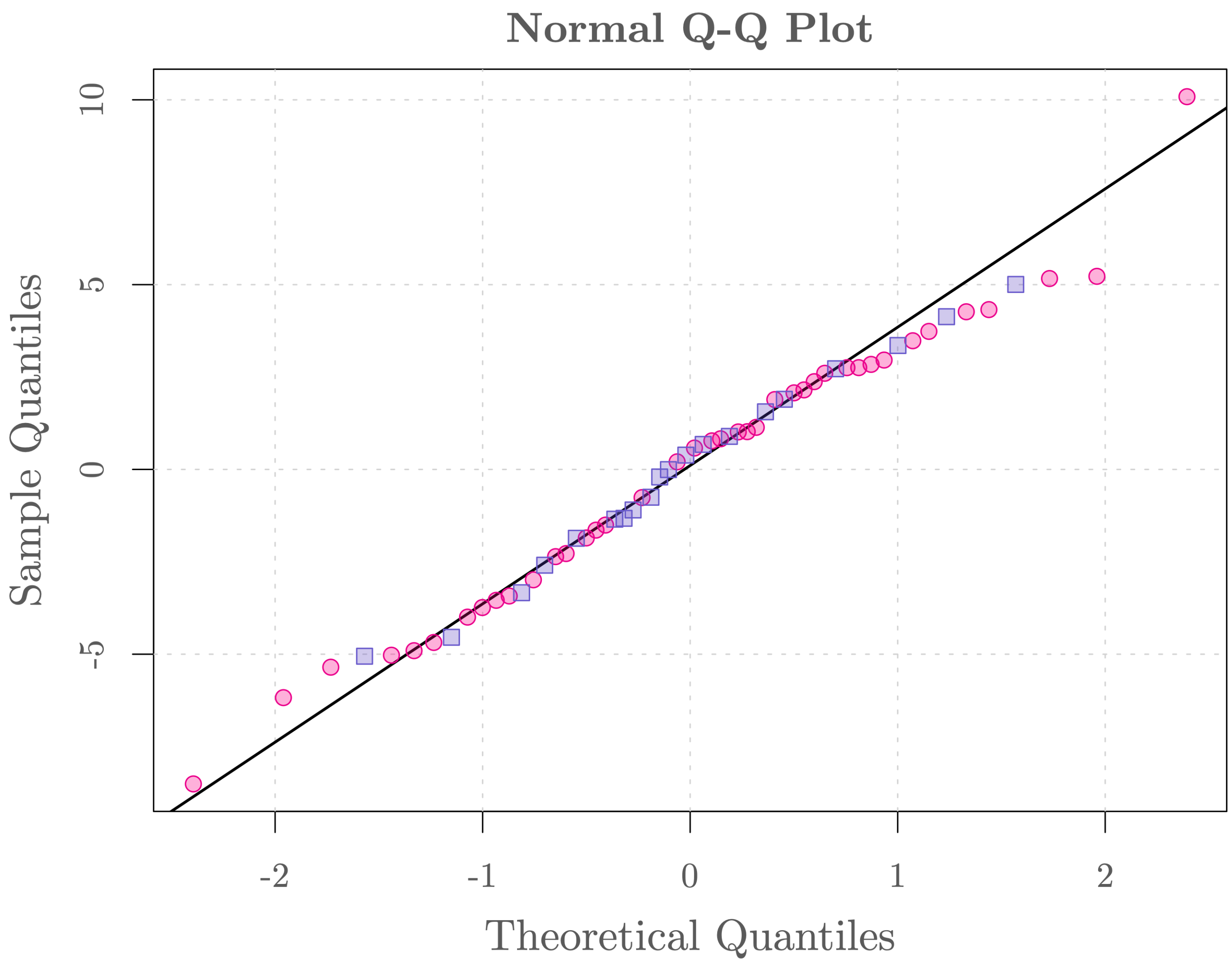

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Concluding remarks

- The last model we consider appears to fit the data, avoids overfitting and allows to answer whether the new reading activities are of interest. Indeed, the programs significantly improve the reading performance of the students (e.g. 5.81 more per year compared to control, p-value < 5%).
- Our model assumes a linear relationship between the response and the covariates. However, this may be incorrect.
- Our model only considers independent data (which may not be correct here).
- Finally, linear regression should not be used to extrapolate, i.e. to estimate beyond the original observation range. For example, if we consider a 100 year-old person in this reading ability example, we would predict (using the third model) that the corresponding score of the DRP test would be 1157.59 and 611.45, respectively, with and without these activities. Does it really make sense? 😯

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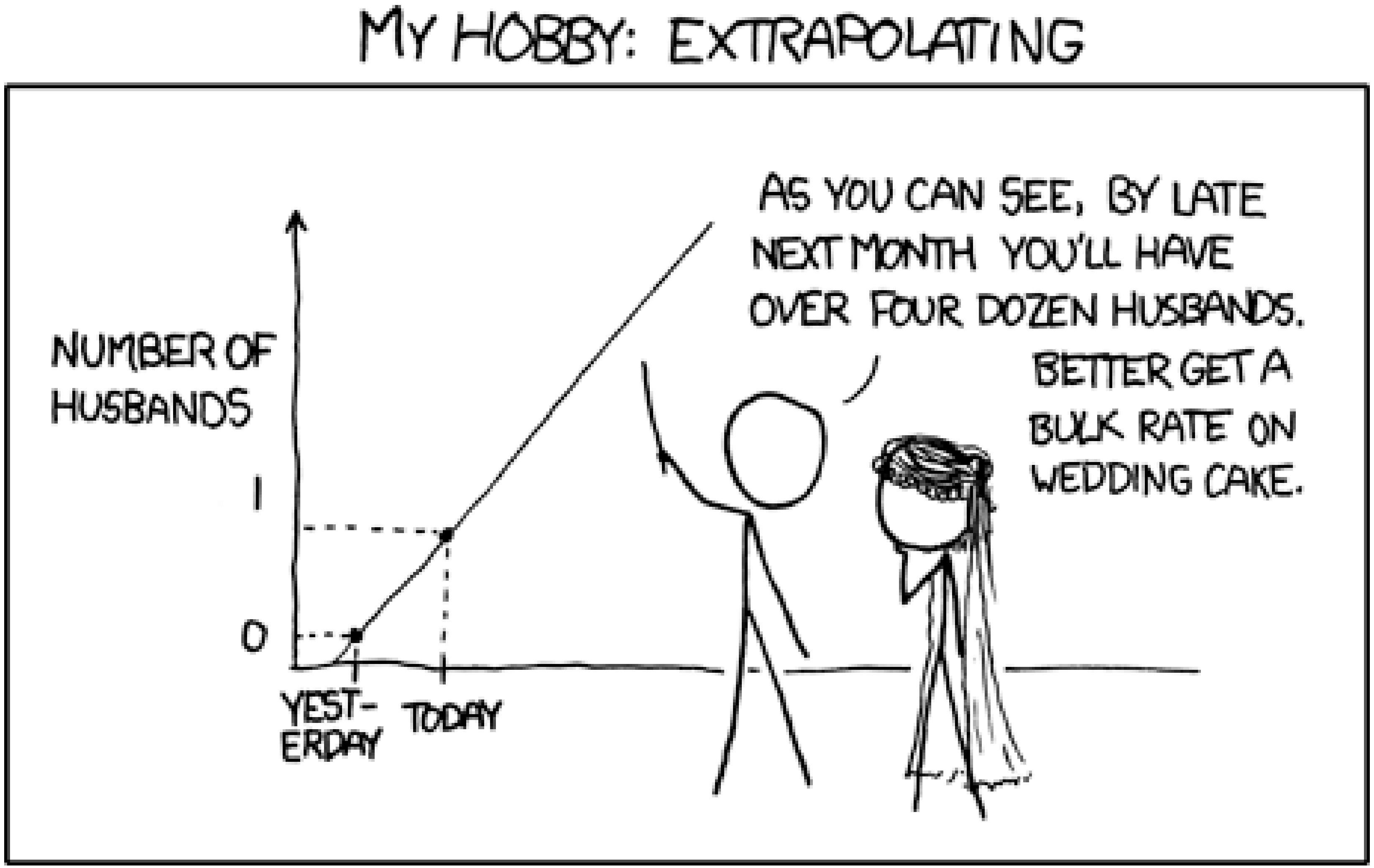
Extrapolating



👋 If you want to know more have a look here.